Accelerating integral decomposition with Kira, machine learning, and exact reductions

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Abstract

We study how to reduce the time required to express loop integrals in terms of master integrals. The method uses exact formulas and recurrences when their mathematical conditions hold, then calls Kira with reduction settings chosen from measured complete costs. Analytic methods cover specified vacuum families and independent products through thirteen loops. We add an explicit Gamma-function evaluation for a massless triangle with two lightlike external momenta, retain its factorized form for independent products, and test native reduction choices on planar boxes, nonplanar boxes, massive sunrise integrals, and a three-loop vacuum family. The existing seed model remains active within its accepted domain. A newly trained cost model and an added equation filter fail their validation conditions and remain disabled. On a matched collection of 93 requests, the final shared-call application takes 41.137 seconds, compared with 45.091 seconds for the preceding shared-call application, reducing time by 8.8 percent. These medians include application loading, exact algebra, output reading, cleanup, and recovery. All 837 application results from the final comparison pass their exact coefficient checks. The gain combines direct formulas, validated native settings, and runtime changes; it does not establish an additional learned-policy gain. The demonstrated range includes selected structures through thirteen loops, while general connected high-loop scattering remains outside the verified domain.

1 Introduction

Feynman integrals arise from integration over internal loop momenta in higher-order perturbative corrections to scattering amplitudes and particle decay amplitudes. Many integrals in a calculation share propagators and differ only in their powers or numerators. Integration-by-parts (IBP) identities relate these integrals and express them in terms of a smaller set of master integrals [1, 2]. This decomposition supplies the coefficients needed to assemble an amplitude; the evaluation of any remaining master integrals is a subsequent calculation [1].

Kira automates the generation and solution of the IBP system, and its current methods already remove unnecessary seeds and equations [3, 4]. The remaining cost depends on the integral structure and the requested targets. Generating fewer useful equations can reduce work, but a restricted system can also fail to reach the required basis and require a larger calculation. We therefore optimize complete reduction time, including prediction, algebra, and recovery. Seed-count reduction alone does not measure this objective.

Our method first identifies structures that admit an exact reduction formula. It applies those formulas to complete requests, individual targets, and recognized intermediate sectors. For the remaining supported native calculations, we retain the seed policy and compare learned selection of symbolic or finite-field algebra with a fixed setting for each propagator structure. Exact algebra determines every output coefficient. Related machine-assisted searches motivate the seed intervention [5, 6], while analytic one-loop reduction motivates the formula stage [7, 8]. We derive the particular formulas and cost objective used here in the appendices.

The method combines exact partial fractions, tensor and scalar recurrences, and analytic values for specific connected trees. We then address the remaining native coefficient algebra with finite-field reconstruction and a trained cost selector. A matched comparison separates the direct formulas, the established reconstruction algorithm, and the added learned decision. Earlier equation-selection and row-selection results remain separate, and the companion notes preserve their full experiments.

2 The decomposition problem and Kira

We first define the notation used in this paper. An integral family specifies inverse propagators, masses, and external scalar products. In dimensional regularization, we write [1, 3]

$$I_{\boldsymbol{a}}(\boldsymbol{x}, d) = \int \prod_{h=1}^{L} \frac{d^d \ell_h}{i\pi^{d/2}} \prod_{j=1}^{N} D_j^{-a_j}, \qquad D_j = Q_j^2 - m_j^2 + i0, \qquad d = 4 - 2\epsilon. \tag{1}$$

Here L is the number of loops, ℓ_h is loop momentum h, N is the number of integral coordinates, and Q_j is a specified linear combination of loop and external momenta. The quantity m_j is the mass on line j, $\boldsymbol{x}$ contains squared masses and external invariants, ϵ is the dimensional regulator, and $i0$ specifies the Feynman pole prescription. A positive integer a_j gives a denominator power, zero removes that factor, and a negative integer gives a numerator power. Auxiliary inverse propagators complete the coordinates for loop scalar products when the physical lines do not suffice [3].

A sector consists of integrals with the same positive-index positions. Removing a positive position gives a sub-sector. A seed $\boldsymbol{\sigma}$ is an index vector at which we generate identities [3]. For a loop or external momentum v, the IBP relation is [2]

$$0 = \int \prod_{h} \frac{d^d \ell_h}{i\pi^{d/2}} \frac{\partial}{\partial \ell_k^\mu} \left(v^\mu \prod_j D_j^{-\sigma_j} \right) = \sum_u A_{\boldsymbol{\sigma},o;u}(\boldsymbol{x}, d) I_{\boldsymbol{a}^{(u)}}. \tag{2}$$

The operator label o specifies (k, v), and $A_{\boldsymbol{\sigma},o;u}$ is the coefficient of integral u. Appendix A expands the derivative into index shifts and explains why the coefficients are rational. Each seed supplies several operators and therefore several equations.

Kira orders integrals, uses symmetries and other valid relations, selects equations, and performs elimination and back-substitution [3, 4, 9]. For a target T, the result has the form

$$T = \sum_{m=1}^{n_M} r_{Tm}(\boldsymbol{x}, d) M_m, \qquad r_{Tm} \in \mathbb{Q}(\boldsymbol{x}, d). \tag{3}$$

Here M_m is master integral m, n_M is the number of masters, and $\mathbb{Q}(\boldsymbol{x}, d)$ is the field of rational functions with rational coefficients. Exact row operations give Eq. (3), as derived in Appendix A. A remaining free integral in a restricted system does not by itself establish a master of the full family. We check completion against a reference basis. Kira can solve symbolically or use finite fields and reconstruct rational functions [10]; this study leaves its reconstruction algorithm unchanged.

The original three learned interventions occur at distinct points (Table 1). A seed policy changes which identities are generated. An equation policy selects seed–operator pairs. A row policy chooses among allowed elimination steps after coefficients exist. All three can change cost without predicting the physical coefficients themselves. The present extension adds a model that selects the coefficient-algebra procedure for the remaining native reductions.

Table 1: Learned decisions studied here. The last column states their use in the measured method.

Decision	Work affected	Use in this study
Seed region	Generation and solution of the IBP system	Retained seed-cost model
Equation retention	Construction of selected IBP coefficients	Separately validated native filter
Elimination row	Intermediate expressions and row operations	Omitted from the retained application

3 Exact reductions before native elimination

The exact stage uses the propagator structure to determine whether a formula applies. For independent loop components, the product measure and separated integrand factorize Eq. (1); Appendix A proves the resulting product of component reductions. Identical components share their calculation. Connected graphs require their own identities, so a high loop count alone does not select this shortcut.

Table 2 summarizes the retained domains. For generic one-loop sectors, we resolve numerator scalar products along the external directions and average the perpendicular components. This yields scalar integrals whose raised powers are lowered by an exact IBP recurrence. Appendix F derives both steps. Their coefficients are rational in d, and keeping known linear dimension factors separate reduces repeated symbolic simplification. A singular matrix can prevent those particular recurrences from being used, which motivates the new two-point rule.

Table 2: Exact identities used by the application. Bounds describe the implementation and do not classify all reducible integrals.

Integral structure	Identity and accepted domain
Generic one-loop sectors	Tensor angular average and scalar IBP recurrence; rational kinematics, at most five coordinates, numerator and excess positive powers at most six; required matrices invertible
Zero-invariant two-point sectors	New endpoint formula; equal, unequal, or zero masses; rational kinematics, at most five coordinates, numerator and excess positive powers at most four
Scaleless vacuum families	Homogeneity gives zero; no masses or external momenta, up to 128 coordinates and thirteen loops
Vacuum factors	Partial fractions and analytic tadpoles; rational nonnegative masses, up to eight denominators per factor, and products through thirteen loops
Symbolic two-point masses	New scalar mass-derivative formula at $p^2 = 0$; individual powers at most six and total power at most eight
Connected one-scale trees	Successive massless bubble integration; massless two-point root or one massive vacuum root; implemented through thirteen loops
One-scale massless triangles	Two lightlike legs and one spacelike invariant; direct Gamma ratios with powers through four and total power at most eight
Independent products	Product of accepted component reductions; repeated components computed once

3.1 Two-point sectors at zero external invariant

Let $D_1 = \ell^2 - x$ and $D_2 = (\ell - p)^2 - y$, where x and y are squared masses and $p^2 = 0$. Write $T(x)$ for the unit-power tadpole with squared mass x. Feynman parametrization gives

$$B_{11}(x, y) = \frac{T(x) - T(y)}{x - y}, \qquad B_{ab}(x, y) = \frac{\partial_x^{a-1} \partial_y^{b-1} B_{11}(x, y)}{(a-1)!(b-1)!}. \tag{4}$$

Here a and b are positive propagator powers, and derivatives treat the two masses as independent before any specialization. Appendix G derives this relation, its equal-mass limit, and the finite tensor formula. At equal masses, $B_{ab}(x, x) = T_{a+b}(x)$, where T_n is the tadpole with denominator power n. A zero-mass tadpole vanishes in dimensional regularization [1].

For polynomial numerators, the same parameter shift leaves one radial denominator. Angular moments reduce the numerator to a finite sum, and a change from the Feynman parameter to the interpolated squared mass expresses each term through the two endpoint tadpoles. The result is a rational coefficient of $T(x)$ and $T(y)$, with no inverse external Gram matrix. This also applies when a generic one-loop recurrence reaches such a two-line sub-sector. Applying the identity at that intermediate step extends the complete reduction beyond requests that begin as bubbles.

The application tests the stated conditions before using a formula. It retains native reduction for unsupported targets, symbolic numerators outside the accepted domain, and recurrence limits. A request can contain both accepted and remaining targets; the application combines their exact outputs. These checks determine mathematical applicability. They do not ask a learned model to certify an identity.

3.2 Direct vacuum evaluation

A massless vacuum family has no dimensional scale and vanishes in dimensional regularization [1]. We apply this check before any elimination, including to coupled loop momenta. Appendix I gives the radial scaling argument and the implemented conditions.

When every denominator in a component carries the same loop momentum, its integrand is a rational function of that momentum squared. Equal masses combine their powers, and partial fractions express the remaining poles through raised-power tadpoles. Appendix I derives the coefficients and evaluates each tadpole through Gamma functions. Polynomial terms and massless vacuum factors vanish in dimensional regularization [1]. The application therefore returns both a master decomposition and an explicit analytic value for this domain.

This rule also handles repeated masses and polynomial numerators. It applies independently to each separated loop factor and computes identical components once. Recognition requires the absence of a momentum coupling between factors. Connected graphs use a separate identity.

3.3 A direct one-scale triangle formula

A massless triangle with two lightlike external momenta admits an explicit Gamma-function value [11]. We use this identity to return both its reduction coefficient and its analytic value. Let T_{abc} have denominators k^2, $(k-p_1)^2$, and $(k-p_1-p_2)^2$, with $p_1^2 = p_2^2 = 0$ and $Q = -(p_1+p_2)^2 > 0$. The integers a, b, c are their powers. With $B(Q, d) = T_{101}$, the unit-power example is

$$T_{111} = \frac{2(d-3)}{Q(d-4)} B(Q, d). \tag{5}$$

Appendix M derives the general coefficient as a finite product of linear factors in d and gives the Gamma-function value of B. The code accepts nonnegative powers through four with $a+b+c \leq 8$, rational $Q > 0$, and no extra occupied denominator. It also treats independent products of accepted triangles through thirteen loops.

The preceding application already returned analytic ratios to a triangle master on these inputs. The new implementation evaluates the coefficient directly in a bubble basis and supplies its explicit analytic value. We compare the two outputs after an exact basis conversion, and we also compare the single-loop results with Kira. The timing comparison therefore measures the change from the preceding analytic application, including its setup, rather than attributing all of the gain to avoided IBP elimination.

3.4 Connected trees and higher loop counts

Successive massless bubble integrations also reduce specific connected trees to a unit-power integral of the same tree. For a massless two-point root, the coefficient is a rational function of d and the external invariant. For a massive vacuum root, it is rational in d and the one squared mass. Appendix H derives these statements from Gamma-function products and their integer shifts. Equation (69) gives the complete analytic value of a connected vacuum tree, and the application now returns this value with its reduction. This family includes connected high-loop integrals, while independent products provide a separate high-loop test. Neither construction represents general coupled multiscale scattering.

4 Learning the remaining Kira cost

Exact formulas remove some reductions from the native calculation. For the remaining integrals, the cost still depends on the seed region and on how the coefficients are represented during elimination. We keep the previously validated seed policy and train a new model to choose the algebra procedure. This separates a change in the equations from a change in the arithmetic used to solve them.

4.1 Seed, equation, and row decisions

The retained seed model chooses among the wide, tight, and medium regions of Appendix B. Its cost tree has maximum depth two, and its failure guard uses twelve depth-three trees per action. The original training used 34 batches and sixteen validation batches. We preserve those weights during the present algebra comparison. The companion notes describe that training and a later unsuccessful replacement using complete seed regions.

The separately tested equation model ranks seed–operator pairs before their full coefficients are constructed. It reduces time on its recorded workload, but the present combined application keeps this filter off. The original next-row model scores allowed elimination steps in the changing matrix. Its additional cost exceeded the algebraic savings in the retained comparison, and a later compiled policy did not establish a consistent saving. We omit learned row selection and retain the native elimination rules. Section 6 gives the separate controls for these three interventions.

4.2 Choosing the algebra procedure

Direct symbolic elimination carries rational functions through each operation. Kira can instead solve the equations at finite-field points and use FireFly to reconstruct the rational functions [10, 12, 13]. The preceding native executable uses symbolic reduction without FireFly support. The new executable retains the same project changes to Kira and adds its standard FireFly capability. We use that reconstruction algorithm without changing its mathematics. Our added decision selects direct symbolic reduction or full reconstruction with a probe batch size of 1, 2, 4, 8, or 16. The probe batch size is the number of numerical evaluations grouped by the native calculation. It differs from the number of target integrals in a request.

We prepare 48 training requests, 22 validation requests, and 22 reserved native requests. They contain planar and nonplanar two-loop boxes, massive two-loop sunrise integrals, and a massive three-loop vacuum family. Box batches contain up to six targets with numerator rank at most four. Additional requests test the sector corner and raised propagator powers. The full physical inputs are disjoint between splits; graph structures and some individual target vectors recur. Appendix K specifies the propagators, targets, kinematics, and split.

Each input has 29 features derived before reduction. They describe loop and external-momentum counts, propagator incidence, masses, sectors, target powers, and the size of the rational kinematic numbers. Family names do not enter the learned features. Appendix C defines the features and training objective. Training measures one complete application per action, and validation measures three. Failed trials retain their elapsed cost and an estimated recovery cost from the measured symbolic control.

We fit ensembles of 32 extremely randomized regression trees [14]. Each ensemble predicts relative complete cost and an observed failure fraction for all six actions. Separate validation selects the depth, minimum leaf population, and required predicted saving. A fixed control also selects its best action separately for each propagator structure. Thus the learned model must improve on a structure-dependent setting. Deployment requires at least three percent less validation cost and no failed selected validation attempt. Inputs outside the measured feature ranges use the preceding application, and an unsuccessful reconstruction falls back to its symbolic procedure.

The model selects a method before Kira generates and solves the equations. It supplies no reduction coefficients. Exact formulas retain their mathematical admission checks, and the native comparison checks every returned rational coefficient against a separate symbolic reduction in the same master basis. Reconstruction and basis admission alone do not constitute a formal proof for every new input. The independent coefficient checks support the tested results.

4.3 A further native cost-model comparison

The new native study tests whether the preferred probe batch size changes with the requested integrals. We measure 48 training requests and 20 validation requests across the same planar box, nonplanar box, sunrise, and three-loop vacuum structures. For the two-loop families, the five actions are the preceding application and full reconstruction with probe batch sizes 1, 4, 8, and 16. One three-loop training comparison failed and could not be reproduced by the subsequent symbolic and reconstruction checks. We therefore retain the preceding application for the vacuum structure and exclude its candidate reconstruction costs from model fitting. The fitted training set contains 36 two-loop requests; the complete comparison retains all four structures. Each action retains the preceding learned seed procedure, exact algebra, and bounded recovery. Complete action times include the initial choice and any recovery.

We fit 32 extremely randomized trees [14] to predict each action's complete time relative to the preceding application. Baseline seconds weight the squared regression error. Validation selects tree depth from 2, 4, and 6, minimum leaf population from 2 and 4, and the required predicted saving from zero, five, and ten percent. The fixed control selects its action separately for each propagator structure. We enable the learned selector only if it saves at least three percent on validation and selects no failed algebra procedure. This condition tests the value of the added model beyond structure-dependent settings. We use the same 29 physical features and weighted cost objective as Appendix C. Here the training cost is the measured complete time, including actual recovery. The model predicts five relative costs; the stated structure guard and observed validation outcomes determine admission.

Summed action costs on validation give 46.438 seconds for the fixed control and 46.495 seconds for the best fitted selector. The additional native cost selector does not pass the validation condition. The application therefore uses the validated fixed settings and retains the fitted model

for inspection.

We also test the preceding nonplanar equation classifier with the new reconstruction setting at its frozen threshold of 0.05. Five validation requests with two repetitions give 9.824 seconds without this additional filter and 16.807 seconds with it, including recovery. The classifier skips 17,808 of 53,430 candidate equation constructions, but all ten filtered attempts fail the prescribed-master-basis check and require recovery. We therefore leave this additional filter disabled; the earlier seed policy remains active.

5 Comparison design

We fix the trained weights, selected settings, procedures, and reserved inputs before the final comparison. The first reconstruction comparison contains 22 new native requests. A separate analytic test contains 53 previously verified requests: thirteen two-mass vacuum products, 34 connected one-mass vacuum trees, three zero-invariant tensor bubbles, and three ordinary one-loop tensor requests. The two vacuum grids cover every loop order from one through thirteen. These analytic requests test preservation of the exact methods and are excluded from the new model's training. The companion notes also retain the preceding 26-request repeated-mass and scaleless coverage test.

In that first comparison, four applications process the same requests. The earlier application precedes the direct vacuum additions. The preceding application includes those formulas and the retained native seed model. The fixed application adds a validated algebra setting for each propagator structure. The integrated application enables the new cost model only if validation admits it. The fixed and integrated applications share all exact formulas, native settings, and preceding seed decisions. Their comparison isolates the additional algebra-selection model. Comparing the integrated and earlier applications measures the cumulative change across these two extensions on one matched workload.

A subsequent paired comparison uses the same 22 reserved native requests to isolate the inherited learned choices. It holds the validated reconstruction settings fixed and compares the retained learned procedure with its existing all-fixed procedure. Both use the same protected bases, exact references, and resource bounds. This comparison performs no additional fitting or selection from reserved outcomes. Its timings are measured together and are reported separately from the first four-method experiment.

Each timed application starts a fresh process. Its elapsed time includes imports, input preparation, model loading and prediction, exact formulas, native equation generation and solution, any recovery, output, and exit. Reference preparation and the external coefficient comparison lie outside this timer. We run one computational thread per application, use three repetitions, and rotate method order on the same worker. An application has a 90-second outer limit; the bounded native runner can impose an earlier limit on an individual attempt. All methods use the same protected master basis where the native reduction requires one.

Vacuum-product references use independent symbolic partial fractions. The connected-tree comparison checks the preceding exact rule, supported by the derivation in Appendix H. Native and tensor references use separate symbolic Kira calculations. Reference preparation records unsuccessful attempts separately from failures inside a timed application. A request counts as complete only if all repetitions agree with the corresponding exact reference. We report failure and completion counts beside time ratios.

The primary time estimator is the ratio of summed per-request median times on the matched completed subset. Appendix E defines it and the paired request-resampling interval. This measures the sampled collection, so results for distinct workloads are not multiplied into a general speed factor. The four-method comparison provides the cumulative result directly.

6 Measured results

The following results come from the frozen comparison defined in Section 5. Table 3 gives completion and elapsed time by physical structure. Each time is a sum of per-request medians on the same completed inputs for both methods.

Table 3: Current application versus the preceding application with direct vacuum formulas. Done gives preceding/integrated completions; paired gives the number completed by both.

Domain	Done	Paired	Preceding (s)	Integrated (s)
Connected trees, 1–13 loops	34/34	34	5.839	5.788
Zero-invariant tensors	3/3	3	0.596	0.602
Ordinary one-loop tensors	3/3	3	0.731	0.728
Vacuum products, 1–13 loops	13/13	13	2.147	2.132
Two-loop planar boxes	9/9	9	18.206	11.360
Two-loop nonplanar boxes	9/9	9	45.732	23.320
Two-loop massive sunrise	2/2	2	2.658	2.439
Three-loop massive vacuum	2/2	2	3.713	3.482
All 75 requests	75/75	75	79.624	49.852

The integrated-to-preceding ratio is 0.626, with central 95% request-resampling interval $[0.485, 0.816]$. Its point estimate corresponds to 37.4% less time. The integrated application is slower on 29 matched requests. These are complete application times, so the reduction includes the cost of loading and choosing the method.

6.1 Native algebra and learned selection

The 22 matched native requests change from 70.310 to 40.602 seconds, a ratio of 0.577 with interval $[0.420, 0.774]$. The selected finite-field reconstruction is an established Kira capability. Its effect must be separated from the model that selects its settings.

Training uses 48 verified requests and validation uses 22. Validation selects maximum tree depth 4, minimum leaf population 2, and saving margin 0.05. The selected model gives 47.615 seconds of validation cost, compared with 48.173 seconds for the structure-dependent fixed control. This does not satisfy the deployment condition, so the integrated application uses the validated fixed settings. The fitted model and its measurements are preserved, and its weights are not loaded for production decisions.

On the reserved native sample, the fixed application takes 40.274 seconds and the integrated application takes 40.602 seconds. Their ratio is 1.008, with interval $[0.999, 1.018]$. Both applications use the same selected settings, so their difference measures run-to-run variation and the shared entry procedure. It is not an added ML speedup. The final repetition changes 0 of 22 actions relative to the fixed control. The feature-range guard sends 3 requests to the preceding symbolic procedure. Median configuration-loading and decision time is 105.779 milliseconds, including the first loading of the feature dependencies. This quantity is not pure forest evaluation time. The preceding seed model remains shared by both applications.

6.2 Cumulative change and exact reductions

The matched comparison across the direct vacuum and reconstruction changes changes from 81.827 to 49.852 seconds on 75 completed requests. Its ratio is 0.609, with interval $[0.470, 0.787]$, or 39.1% less time at the point estimate. This result uses one input set and one timing procedure. It is a cumulative comparison with the earlier application, not with unmodified Kira, and it does not multiply percentages from older experiments.

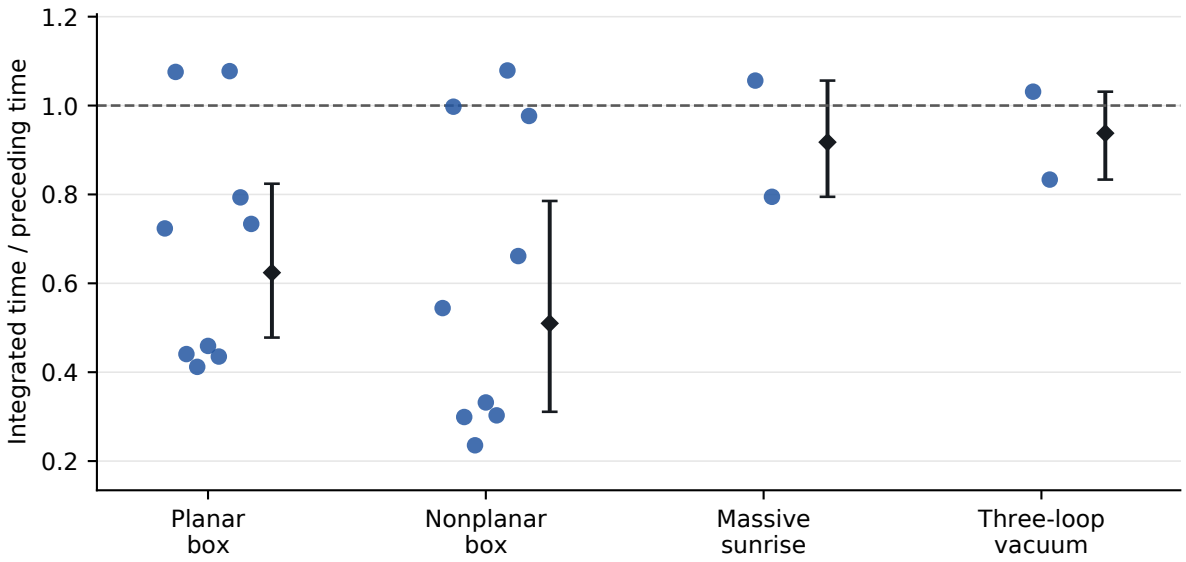


Figure 1: Native speed changes depend on the physical request. Each circle shows the integrated-to-preceding ratio for one completed request. Diamonds show ratios of summed median times within each structure, with paired request-resampling intervals. Values below one mean less elapsed time.

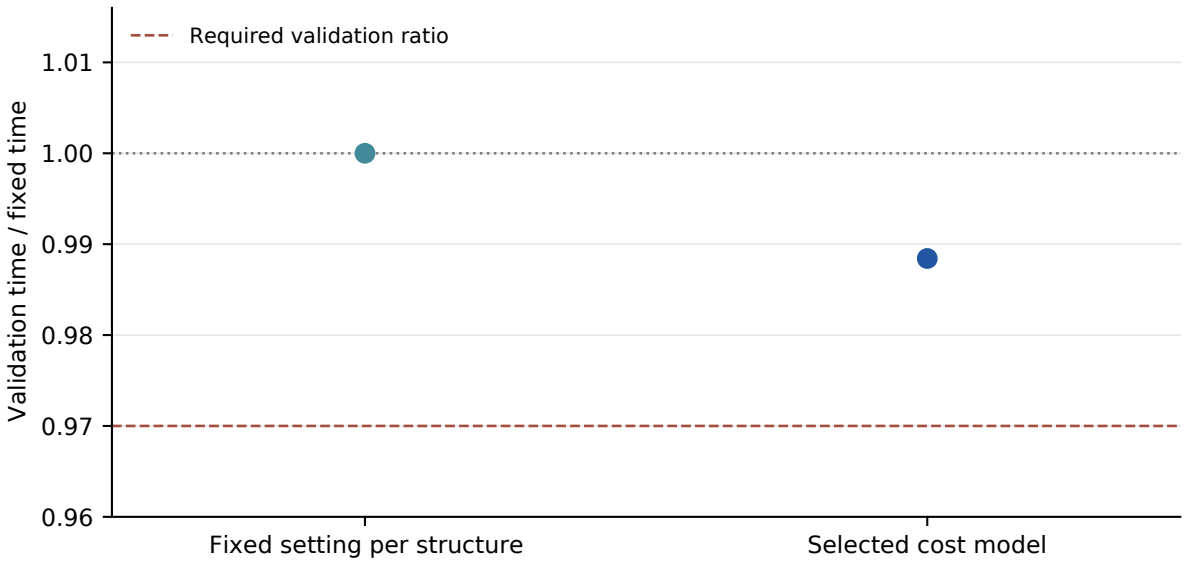


Figure 2: Separate validation determines whether the new model is used. The dashed threshold requires at least three percent less cost than the fixed setting selected for each structure. The current deployment retains the fixed settings because the model does not meet this condition.

The direct vacuum formulas and connected-tree recurrences remain active through thirteen loops. They provide rational master decompositions and explicit Gamma-function values in the domains derived in the appendices. The present analytic grid verifies their preservation while the new algebra method changes the native calculations. General coupled multiscale scattering through thirteen loops is outside this grid.

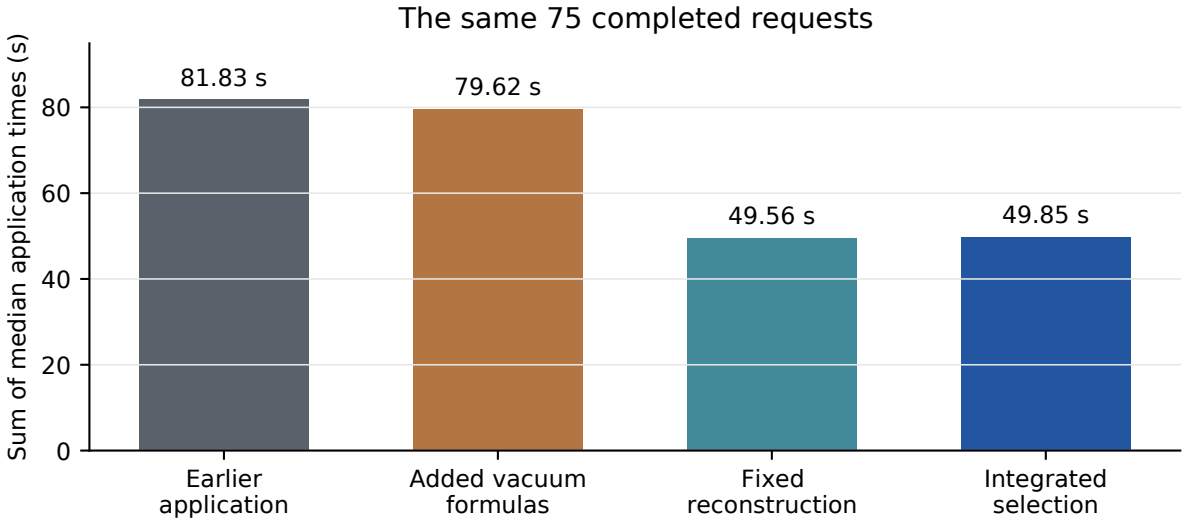


Figure 3: Cumulative elapsed time is measured on the same requests for all four applications. The second application adds direct vacuum formulas; the third adds validated fixed reconstruction settings; the fourth applies the admitted selection procedure. The bars do not combine unrelated historical speed factors.

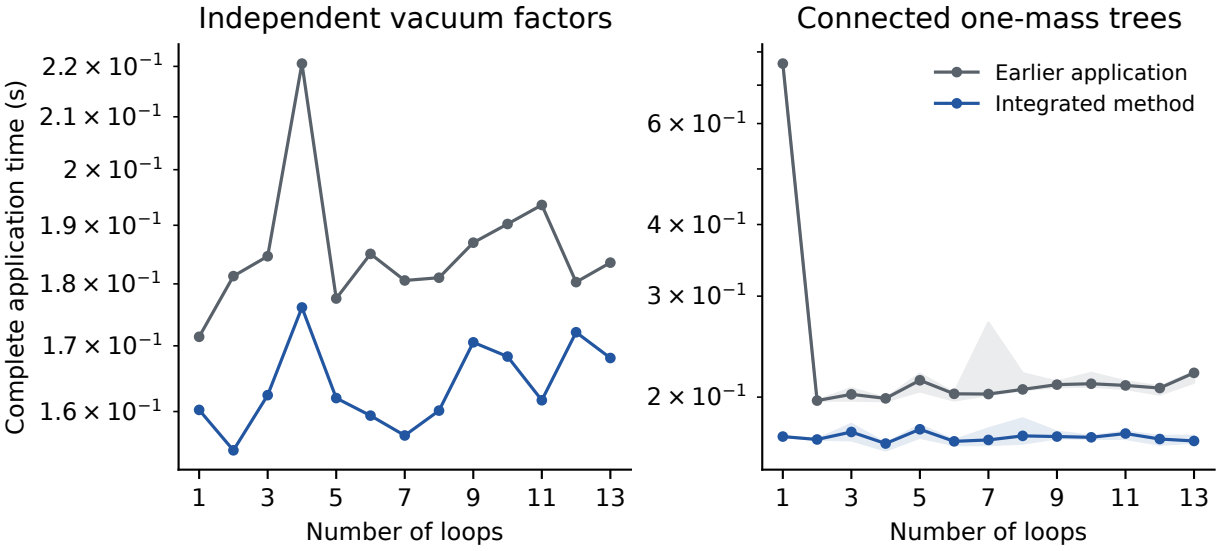


Figure 4: The analytic stage covers each loop order from one through thirteen in its stated domains. The left panel shows independent vacuum factors. The right panel shows connected one-mass trees, with the median over available shapes and their full range. Logarithmic time axes show the small complete application times without hiding their variation.

6.3 Completion and scope

The timed outputs receive 18,132 exact coefficient comparisons. The integrated application has 0 unverified timed applications and 0 recorded unsuccessful inner native attempts. It uses reconstruction recovery 0 times. Reference preparation fails for 0 final requests. Application failure

counts include a terminated run even when it cannot return an inner-attempt record.

The current native test measures new kinematic points and complete target batches within four known structures at two and three loops. Higher-loop evidence comes from the derived product and tree identities. It does not establish performance for every high-loop graph, every symbolic kinematic system, or unrestricted numerator powers. The measured configuration also depends on the tested processor and resource bounds. A finite model and setting search does not prove globally minimum time.

6.4 Retained ML in the reconstruction workflow

The additional paired comparison holds the new algebra settings fixed and switches the inherited learned choices to their existing fixed procedure. On 22 matched native requests, the fixed procedure takes 42.175 seconds and the learned procedure takes 41.448 seconds. Their ratio is 0.983, with interval $[0.951, 1.003]$. This interval includes equal time, so the sample does not resolve a consistent additional learned saving. The recorded seed-attempt sequences differ on 14 requests. The learned and fixed procedures have 0 and 0 unverified applications, respectively. Across all three repetitions, their total times are 128.032 and 127.546 seconds. The distinction between total repeated time and summed medians illustrates the timing variation. This ablation has its own paired timings; its ratio is not multiplied into the cumulative result.

Table 4: Inherited learned choices versus their existing fixed procedure, with the new reconstruction settings shared. The columns use matched completed requests within each structure.

Structure	Paired	Fixed (s)	Learned (s)	Ratio
Planar box	9	11.656	11.482	0.985
Nonplanar box	9	24.355	23.842	0.979
Massive sunrise	2	2.565	2.542	0.991
Three-loop vacuum	2	3.600	3.582	0.995
All native requests	22	42.175	41.448	0.983

6.5 The other learned interventions

Table 5 retains the main earlier controls needed to interpret the method. They use different workloads and cannot be multiplied into a cumulative speed factor. The equation-filter result supports that intervention on its tested domain, while the row results motivate its omission. The current combined experiment keeps equation filtering off and does not claim a joint seed-plus-equation gain. The earlier seed comparison has interval $[0.937, 0.984]$, and the equation

Table 5: Earlier measurements from this study, retained as separate evidence. A ratio below one means less time. Their full input definitions and measurements remain in the notes.

Intervention	Control and workload	Time ratio
Earlier native seed model	Fixed region, 32 box batches: 156.383 to 150.663 s	0.963
Equation selection	Native filter, 24 new-index batches: 72.295 to 67.478 s	0.933
Original learned row model	Native row choices, 84 trials	1.305
Compiled row-cost policy	Native row choices; geometric mean	0.982

comparison has interval $[0.928, 0.940]$. The compiled row policy has interval $[0.959, 1.003]$, which includes equal time. These intervals use the resampling procedure recorded for each original study. The native row algorithm remains in the retained workflow.

6.6 Processing a collection together

The preceding comparison loads the application separately for each request. A further test processes the same 75 requests in one call, so numerical routines, mathematical objects, and learned policies can remain in memory. The shared call also uses one bounded output stream for the collection. Every request still receives its own reduction, and the exact formulas, model weights, algebra settings, and protected bases remain unchanged. The input supplies no reference coefficients.

We compare separate and shared calls for both the earlier and current methods. The earlier method precedes the direct vacuum formulas and reconstruction extension. Each comparison uses three complete repetitions. Two workers run on the same Intel Gold compute node, with one computation thread per application. We alternate method order and reverse the request order in the middle repetition. Time includes loading, all reductions, output, exit, reading the returned data, and cleanup; the separate-call total sums those complete call times. Independent reference comparisons remain outside both timers. These implementation checks reuse the previously verified physical inputs.

Table 6: Complete time for the same collection of 75 requests. Each value is the median of three full-collection times. The range contains all three repetitions and is not a confidence interval. Every method completes every request in every repetition.

Method	Median (s)	Observed range (s)
Earlier method, separate calls	96.355	94.794–103.077
Current method, separate calls	58.980	57.669–59.163
Earlier method, shared call	70.051	69.210–71.538
Current method, shared call	37.247	36.713–38.256

For the current method, shared processing changes complete time from 58.980 to 37.247 seconds, a ratio of 0.632. The matched cumulative comparison changes from 96.355 seconds for the earlier separate calls to 37.247 seconds for the current shared call. When both methods share their setup, the earlier method takes 70.051 seconds and the current method takes 37.247 seconds. This control preserves a comparison of the algebraic changes after both methods receive the same setup benefit. These collection-level medians differ from the sum of per-request medians in Table 3 and come from a separate timing job. We compare each method with the controls measured in the same experiment.

All 900 request executions pass 18,132 coefficient comparisons, with zero failed applications. Shared processing preserves every recorded reconstruction choice. This change comes from reuse within a call. The models and verified physical domain remain those of the preceding comparison.

6.7 Direct triangles and refined native choices

We freeze the added formula and cost policy before testing 40 new requests. Twenty contain native two- or three-loop reductions, and twenty contain single triangles or independent triangle products. The product tests cover every loop order from one through thirteen. We measure three fresh-process repetitions for each request and method. These first measurements retain the original runtime monitor. The preceding application, the new fixed control, and the new integrated method receive the same physical inputs. Times include application loading, model use where enabled, algebra, output reading, and cleanup. Independent coefficient comparisons remain outside all timers.

The native requests give a total-time ratio of 0.887, with a paired request-bootstrap interval of $[0.743, 1.010]$. The integrated-to-fixed ratio is 1.010, with interval $[0.990, 1.029]$. Both new methods use the same fixed choices because the added selector is disabled. Their timing difference

Table 7: New reserved requests. Times sum per-request medians of three complete calls. The ratio divides new time by preceding time.

Requests	Number	Preceding (s)	New (s)	Ratio
One-scale triangles and products	20	5.231	4.269	0.816
Planar boxes	5	10.829	8.792	0.812
Nonplanar boxes	5	17.214	13.123	0.762
Massive sunrise	5	10.215	10.265	1.005
Three-loop vacuum	5	14.277	14.399	1.009

does not measure a new ML gain. The seven-loop product request has a time ratio of 1.077 in this fresh-process comparison. The measurements therefore include a slower triangle case, and they do not establish a monotonic trend with loop order.

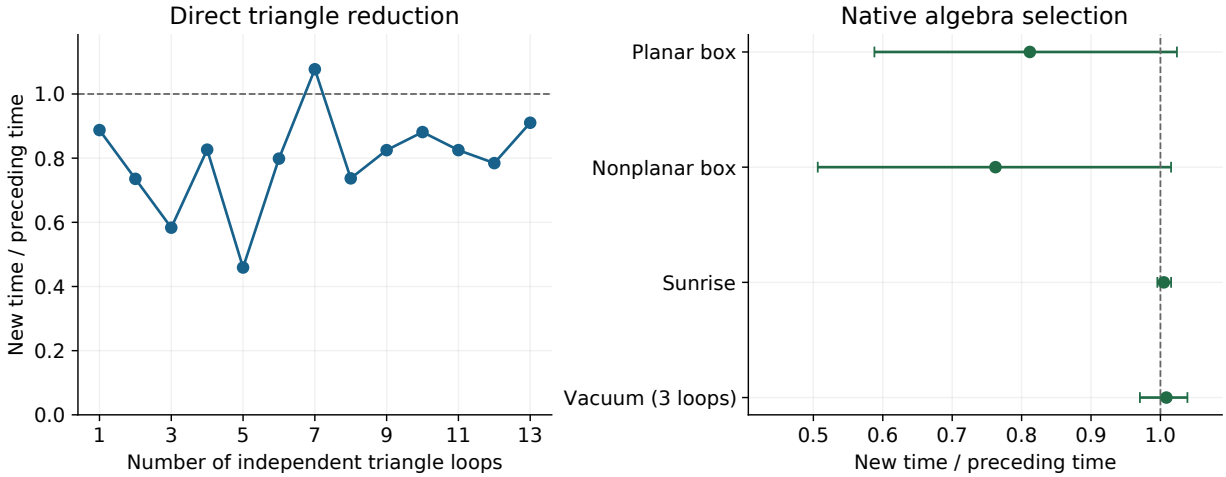


Figure 5: Complete-time ratios on the new requests. The dashed line marks equal time. The left panel shows the direct triangle implementation at each tested product loop order. The right panel shows native requests by propagator structure, with paired request-bootstrap intervals. Product tests do not establish connected higher-loop scattering performance.

6.8 Final complete-call comparison

A native reduction removes temporary files while it runs. The earlier resource monitor could stop a reduction if a directory disappeared during its inspection. We change the monitor to tolerate removal while preserving its storage bound. A controlled check reproduces the old exception and verifies the new behavior. This change affects runtime supervision and leaves the algebra and learned decisions unchanged.

We compare the preceding published application, the new mathematical method with the original monitor, and the final application with the corrected monitor on the same 93 requests. Each method runs three times on the same compute node with one worker, rotating method order, and reversed request order in the middle repetition. A preceding two-worker comparison showed timing differences larger than the measured monitor cost. It also stopped when recovery in the preceding application returned a 20-integral basis instead of the 16-integral reference basis for one vacuum request. We retain that failed comparison separately. The complete serial check uses every planned request and method. The complete timer includes unsuccessful internal attempts and recovery. Independent comparisons check the saved exact coefficients outside the timer. The final-to-preceding median ratio is 0.912, which reduces time by 8.8 percent on this collection. The final-to-original-monitor ratio is 0.958. All 837 application results pass 13,401 exact coefficient

Table 8: Final matched shared-call comparison. Times are complete collection times. The range shows the three observations. Internal failures count attempts across all three repetitions.

Method	Median (s)	Mean (s)	Range (s)	Failures
Preceding application	45.091	45.230	44.783–45.817	0
New method, original monitor	42.932	42.624	40.351–44.589	0
Final application	41.137	41.029	40.723–41.227	0

comparisons. This finite timing comparison does not prove a universal speedup or a globally shortest reduction.

7 Conclusion

The additional matched collection gives 45.091 seconds for the preceding shared-call method and 41.137 seconds for the new method. Direct triangle coefficients and validated native settings produce this result within the stated domains. The additional native cost selector does not pass the validation condition. The application therefore uses the validated fixed settings and retains the fitted model for inspection.

In the earlier comparison of 75 matched requests, the integrated method reduces summed complete time by 37.4% relative to its preceding application. The matched cumulative comparison across the direct vacuum and reconstruction changes gives 39.1% less time. Exact partial fractions and recurrences handle the stated analytic families, and finite-field reconstruction reduces the remaining native algebra where the measured settings support it.

For those 75 requests, processing the collection in one call reduces repeated setup. The measured collection time changes from 58.980 to 37.247 seconds, while the equally prepared earlier method takes 70.051 seconds in one call. Every request passes its independent coefficient comparison. This implementation gain is separate from the learned decisions and applies to the tested collection.

The first reconstruction cost model fails its deployment condition, so the production method retains the validated fixed reconstruction settings. The paired ablation of inherited learned choices gives a ratio of 0.983 with interval $[0.951, 1.003]$. Earlier equation-selection measurements remain separate, and learned row selection is omitted. This separation identifies which gains come from analytic identities, established algebra methods, and learned decisions.

The formulas apply under explicit mathematical conditions, and the tests cover selected structures through thirteen loops. New native inputs are restricted to the four stated two- and three-loop families with rational kinematics and symbolic dimension. These results support the reported reduction times within that domain; they do not establish a fastest method for all integrals.

A IBP identities, row operations, and exactness

The starting total-derivative identity follows from integration by parts in a region where the integral converges, followed by dimensional analytic continuation [2, 1]. Write $P_{\boldsymbol{\sigma}} = \prod_i D_i^{-\sigma_i}$ and $\partial_k = \partial/\partial\ell_k$. The product rule gives

$$\partial_k \cdot (v P_{\boldsymbol{\sigma}}) = (\partial_k \cdot v) P_{\boldsymbol{\sigma}} - \sum_i \sigma_i (v \cdot \partial_k D_i) D_i^{-\sigma_i - 1} \prod_{j \neq i} D_j^{-\sigma_j}. \tag{6}$$

For $v = \ell_h$, the first factor is $d\delta_{kh}$; for an external momentum it is zero. Completeness of the inverse-propagator coordinates permits us to write $v \cdot \partial_k D_i = u_i + \sum_j v_{ij} D_j$, where u_i and v_{ij} are coefficients fixed by the propagators and kinematics, with the invariant and mass terms included

in u_i. Integrating Eq. (6) therefore gives

$$0 = (\partial_k \cdot v) I_{\boldsymbol{\sigma}} - \sum_i \sigma_i u_i I_{\boldsymbol{\sigma}+\boldsymbol{e}_i} - \sum_{i,j} \sigma_i v_{ij} I_{\boldsymbol{\sigma}+\boldsymbol{e}_i-\boldsymbol{e}_j}. \tag{7}$$

The unit vector $\boldsymbol{e}_i$ raises index i by one. This derivation explains how a seed and an operator generate a row of the reduction system.

Let q be a source row and k a pivot column with $A_{qk} \neq 0$. Solving the identity $\sum_j A_{qj} I_j = 0$ for I_k gives $I_k = -\sum_{j \neq k} (A_{qj}/A_{qk}) I_j$. Subtracting A_{rk}/A_{qk} times that zero identity from row r yields

$$A'_{rj} = A_{rj} - \frac{A_{rk}}{A_{qk}} A_{qj}, \qquad A'_{rk} = 0. \tag{8}$$

Every step preserves the represented relation. Repetition until only prescribed master columns remain gives Eq. (3). Removing seeds or equations preserves the validity of the retained identities, but can prevent this elimination from finishing. We therefore require reduction to the prescribed basis and exact agreement with the reference coefficients. Generic kinematics avoid singular pivot denominators; special kinematic limits require their own treatment.

For independent components, write the loop momenta as disjoint groups and the integrand as $\prod_b f_b$. In a convergence domain, repeated integration gives $I = \prod_b \int f_b$. Dimensional analytic continuation preserves that identity [1]. Substituting each exact component reduction therefore gives

$$I = \prod_b \left(\sum_m r_{bm} M_{bm} \right). \tag{9}$$

Our product output retains this factored form. It avoids expanding the Cartesian product of master terms and uses the same representation in all timed methods.

B The seed regions and equation score

For seed $\boldsymbol{\sigma}$, define excess positive power $D = \sum_i \max(\sigma_i - 1, 0)$, numerator rank $R = \sum_i \max(-\sigma_i, 0)$, and positive sector $P = \{i : \sigma_i > 0\}$. Let h be the number of absent physical lines relative to the largest top sector, and let D_T, R_T be the largest target values. The distance from the target numerator rectangles is defined by

$$q(\boldsymbol{\sigma}, \mathcal{T}) = \min_{\boldsymbol{a} \in \mathcal{T}} \sum_{i \notin P} \max\{0, -\sigma_i - \max(0, -a_i)\}. \tag{10}$$

Projection onto the interval from zero to each target numerator power leaves precisely this positive excess. Summing the distances and minimizing over targets gives Eq. (10). We define the region by

$$\begin{aligned} q &\leq \max\{0, \lfloor \theta_0 + \theta_1 h + \theta_2 D + \theta_6 R \rfloor\}, \\ R &\leq \max\{0, \lfloor R_T + \theta_3 - \theta_4 h \rfloor\}, \qquad D \leq \max\{0, \lfloor D_T + \theta_5 \rfloor\}. \end{aligned} \tag{11}$$

These inequalities specify a search family rather than a sufficiency theorem. Native outer bounds, retained sector corners, and admissible target and master seeds also apply. The previously studied sparse choices use

$$\begin{aligned} \boldsymbol{\theta}_{\text{wide}} &= (2.5, 0, 0, 1.25, 0, 1, -0.25), \\ \boldsymbol{\theta}_{\text{tight}} &= (1, 0, 0, 0, 1, 1, 0), \\ \boldsymbol{\theta}_{\text{medium}} &= (1.5, 0, 0, 0.5, 0, 1, 0). \end{aligned} \tag{12}$$

The retained native model uses these three sparse choices. The rejected replacement uses the wide and tight choices and a third action that retains all seeds within the tight procedure's native

outer bounds. The native procedure retains its subsector truncation and recovery rules. The model selects one complete procedure before generating equations.

The separately tested equation classifier has seventeen features: four family and request dimensions; three seed counts; two differences from target power maxima; three sector descriptors; exact target membership and minimum index distance; the operator index; and the two target maxima. Positive labels mark equations retained by a complete reference reduction. Sampling at most 2,000 records of each class y in batch c gives each sampled record weight $N_{cy}/(n_{cy}N_c)$, where N_{cy} and n_{cy} are the native and sampled class counts and N_c is the full batch count. Multiplying this weight by n_{cy} preserves the class fraction N_{cy}/N_c. A leaf with weighted totals W_0, W_1 returns $p = W_1/(W_0 + W_1)$, and a fixed threshold retains equations. The decision-tree construction follows Ref. [15]. This empirical score does not certify sufficiency or probability calibration; the native completion check remains necessary.

C Measured-cost learning

For training request i and action a, let t_{ia} be measured elapsed time, f_{ia} the fraction of unsuccessful repetitions, and b_i the measured preceding-application time. We define the training cost and relative cost by

$$C_{ia} = t_{ia} + f_{ia} b_i, \qquad y_{ia} = C_{ia}/b_i. \tag{13}$$

The first expression adds the measured failed attempt and the estimated cost of recovery. The second expresses this cost relative to the same request's control. These are training definitions. The reserved comparison measures actual complete applications, including actual recovery.

The ensemble predicts the vector $\boldsymbol{z}_i = (y_{i1}, \ldots, y_{i6}, f_{i1}, \ldots, f_{i6})$. It minimizes weighted squared error at its regression leaves, following the extremely randomized tree construction of Ref. [14]. For one component z and a leaf containing requests S, its loss is $\sum_{i \in S} b_i (z_i - \mu)^2$. Differentiation with respect to μ gives

$$0 = 2 \sum_{i \in S} b_i (\mu - z_i), \qquad \widehat{z}(S) = \frac{\sum_{i \in S} b_i z_i}{\sum_{i \in S} b_i}. \tag{14}$$

For a cost component, inserting $z_i = C_{ia}/b_i$ gives

$$\widehat{y}_a(S) = \frac{\sum_{i \in S} C_{ia}}{\sum_{i \in S} b_i}, \qquad \arg\min_a \widehat{y}_a(S) = \arg\min_a \sum_{i \in S} C_{ia}. \tag{15}$$

The denominator is independent of the action, which proves the second equality. Thus a leaf ranks actions by their aggregate measured cost on its requests. The ensemble averages the 32 leaf predictions. Weighting by b_i gives slower baseline requests more influence, while relative costs permit one model to represent different physical structures. This local cost interpretation does not prove global minimum time on unseen requests.

Validation compares maximum depths 2, 4, and 6, minimum leaf populations 2 and 4, and predicted saving margins $\eta = 0, 0.05, 0.10, 0.20$. For the fixed action b selected for a propagator structure, we choose the least predicted relative cost among actions with predicted failure fraction at most 0.02. We use that action a_* only if

$$\widehat{y}_{a_*} < (1 - \eta)\widehat{y}_b. \tag{16}$$

This inequality is the defined fractional saving condition. If no alternative satisfies it, the action remains b. A feature outside the training minimum or maximum uses the preceding procedure. Validation must then show a complete-cost ratio below 0.97 with no failed selected attempt before deployment enables the model. Failure scores describe observed training outcomes; a zero score does not certify an unseen reduction.

To specify the features, define target excess power $D = \sum_j \max(a_j - 1, 0)$, numerator rank $R = \sum_j \max(-a_j, 0)$, and line count $P_T = \sum_j \mathbf{1}_{a_j>0}$. A line is incident on a loop momentum when that momentum occurs in its inverse propagator. The 29 features are listed in Table 9. Every feature follows from the supplied indices and propagator definitions. The ordering within

Table 9: Inputs to the algebra-cost model. Means and extrema over targets use the complete requested batch.

Positions	Quantities
1–6	Number of loops, external momenta, integral coordinates, physical propagators, symbolic external invariants, and $\log(1 + n_T)$ for n_T targets
7–13	Maximum and mean D, maximum and mean R, minimum and maximum P_T, and number of distinct sectors
14–18	Number of massive coordinates; number of physical lines containing several loops; maximum loop incidence per physical line; maximum physical-line incidence per loop; mean external-momentum incidence per coordinate
19–22	Minimum D, minimum R, mean P_T, and maximum absolute index
23–25	Mean excess power on massive coordinates; mean numerator rank weighted by loop incidence; mean excess power weighted by loop incidence
26–29	Maximum sum of positive powers; maximum number of absent physical lines; maximum numerator and denominator bit lengths among rational masses and scalar products

positions 7–10 is maximum D, maximum R, mean D, mean R. The remaining entries follow the listed order. The bit lengths describe the supplied rational numbers and do not encode a family identifier.

D Physical test inputs

All profiles below define squared quantities in a common fixed unit. The zero-invariant families use shifts $(0, p, q, r)$ and masses $(x, y, 0, 0)$, with

$$\begin{aligned} &p^2 = 0, \quad p \cdot q = -h - 1, \quad p \cdot r = -h - 2, \\ &q^2 = -2h - 7, \quad q \cdot r = -h - 3, \quad r^2 = -3h - 13. \end{aligned} \tag{17}$$

The unequal case has $(x, y) = (h+3, h+11)$, the equal case $(h+3, h+3)$, the mixed case $(h+3, 0)$, and the massless case $(0, 0)$. Targets have the form $(1 + \delta_1, 1 + \delta_2, -b_1, -b_2)$, where $\delta_1 + \delta_2$ is the excess positive power and $b_1 + b_2$ is numerator rank.

Ordinary tensor families have P physical lines, one auxiliary coordinate, shifts $q_1 = 0$, $q_{i+1} = p_i$, and

$$m_i^2 = h+11i^2 \ (i \le P), \quad m_{P+1}^2 = 0, \quad p_i^2 = -(10h+17i), \quad p_i \cdot p_j = -(h+i+2j-2) \ (i < j). \tag{18}$$

Appendix J specifies the current profiles, targets, and separate vacuum grids.

For the native planar box, let p_1, p_2, p_3 be massless, $2p_1 \cdot p_2 = s$, $2p_1 \cdot p_3 = t$, and $2p_2 \cdot p_3 = -s-t$. The seven physical line momenta are

$$(\ell_1, \ell_1 - p_1, \ell_1 - p_1 - p_2, \ell_2, \ell_2 - p_1 - p_2, \ell_2 - p_1 - p_2 - p_3, \ell_1 - \ell_2), \tag{19}$$

and the auxiliary momenta are $(\ell_1 - p_3, \ell_2 - p_1)$. The nonplanar routing follows Ref. [3]:

$$\begin{aligned} (Q_1, \ldots, Q_9) = (&\ell_1, \ell_2, p_1 - \ell_1, p_2 - \ell_2, p_1 + p_2 - p_3 - \ell_1, \\ &p_1 - \ell_1 + \ell_2, p_1 - p_3 - \ell_1 + \ell_2, \ell_1 - p_3, \ell_2 - p_1 - p_2). \end{aligned} \tag{20}$$

The first seven entries are physical and all masses vanish. Its convention gives $p_i^2 = (p_1 + p_2 - p_3)^2 = 0$, $s = (p_1 + p_2)^2$, and $t = (p_1 - p_3)^2$. Expanding the squares gives $2p_1 \cdot p_3 = -t$ and

$2p_2 \cdot p_3 = s + t$. The low-rank profiles use $(s, t) = (-1009 - 12h, -1021 - 14h)$ for the planar graph and $(-113-18h, -127-10h)$ for the nonplanar graph. Appendix K gives the independently chosen high-rank invariants.

Every high-rank box target is $(1, 1, 1, 1, 1, 1, 1, -u, -v)$ with $u, v \geq 0$ and $1 \leq u + v \leq 4$. The current split uses the profiles and batch counts in Appendix K. Both graph types occur in all splits, so the test measures new batches and kinematic points within these structures.

E Time ratios and uncertainty

Let T_{imr} be complete elapsed time for request i, method m, and repetition r. For the set $\mathcal{C}$ completed by both methods in all repetitions, define

$$S_m = \sum_{i \in \mathcal{C}} \text{median}_{r=1,2,3}\, T_{imr}, \qquad \rho = \frac{S_{\text{new}}}{S_{\text{control}}}, \qquad \text{fractional saving} = 1 - \rho. \tag{21}$$

These definitions compare identical completed requests. Dividing both times by the control time gives ρ, and subtracting gives the fractional saving. Completion counts remain separate because an unfinished reduction has no verified finite speedup. To estimate variation across requests, we draw $|\mathcal{C}|$ paired request indices with replacement and recompute ρ. We report the central 95% of 5,000 such ratios, using the bootstrap construction of Ref. [16]. This interval describes the sampled request collection and does not establish performance for every integral family.

F Exact reduction of polynomial numerators

F.1 Resolving the loop momentum

One-loop tensor reduction expresses loop-momentum numerators through scalar integrals. Established methods include tensor decompositions and generating functions for arbitrary numerator and denominator powers [7, 8]. We derive the particular identities used in our extension here. The algebra uses rational kinematic values and keeps d symbolic. It applies to a one-loop sector and to independent products of such sectors. Applying it inside a coupled graph would require coefficients that depend on the other loop momenta, which this implementation does not support.

Write $D_i = (\ell - q_i)^2 - m_i^2$ and let $S = \{i : a_i > 0\}$ contain the denominator positions. The q_i in this appendix are fixed external shifts; the infinitesimal prescription is understood. Choose $r \in S$, set $k = \ell - q_r$, and define $R_i = q_i - q_r$. The vectors R_i with $i \in S \setminus \{r\}$ span the external directions seen by the denominators. Their Gram matrix and their contractions with k are

$$G_{ij} = R_i \cdot R_j, \qquad u_i = k \cdot R_i = \frac{D_r - D_i + m_r^2 - m_i^2 + R_i^2}{2}. \tag{22}$$

The second equality follows by expanding $D_i = (k - R_i)^2 - m_i^2$ and subtracting it from $D_r = k^2 - m_r^2$. Assume that G is invertible, and split k into its projection onto this span and its perpendicular part:

$$k_\parallel^\mu = \sum_{i,j \in S \setminus \{r\}} R_i^\mu (G^{-1})_{ij} u_j, \qquad k_\perp = k - k_\parallel. \tag{23}$$

Contracting with each R_i gives $k_\perp \cdot R_i = 0$. Contracting the parallel part with itself gives $k_\parallel^2 = u^T G^{-1} u$, so

$$T(D) := k_\perp^2 = D_r + m_r^2 - u^T G^{-1} u. \tag{24}$$

Empty matrices and sums give zero when S has one element.

An inverse propagator D_j with $a_j < 0$ appears in the numerator and can be resolved into the same components. Define $v_{ji} = R_j \cdot R_i$, $R_{j,\perp} = R_j - \sum_{i,h} R_i (G^{-1})_{ih} v_{jh}$, and $z_j = k_\perp \cdot R_{j,\perp}$. Direct substitution gives

$$D_j = B_j(D) - 2z_j, \tag{25}$$

$$B_j(D) = D_r + m_r^2 + R_j^2 - m_j^2 - 2v_j^T G^{-1} u, \tag{26}$$

$$H_{jh} := R_{j,\perp} \cdot R_{h,\perp} = R_j \cdot R_h - v_j^T G^{-1} v_h. \tag{27}$$

Thus the remaining task is to average products of the z_j over perpendicular directions. All positive-power denominators depend on $k_\perp$ only through $T(D)$, so this angular average leaves the denominator unchanged.

F.2 Angular moments and the finite numerator expansion

Let $\nu = d - |S| + 1$ be the dimension perpendicular to the denominator span. Rotational symmetry makes odd angular moments vanish. For $2h$ perpendicular components, the invariant symmetric tensor is the sum of all pairings of metric tensors, multiplied by a coefficient $C_h T^h$. Tracing two indices gives

$$C_h(\nu + 2h - 2) = C_{h-1}, \qquad C_0 = 1, \qquad C_h = \frac{1}{\prod_{b=0}^{h-1}(\nu + 2b)}. \tag{28}$$

The trace factor contains ν from the pairing of the traced indices and $2h - 2$ from their pairings with the other indices. Contracting the tensor with the external perpendicular vectors therefore gives

$$\left\langle \prod_{t=1}^{2h} z_{j_t} \right\rangle = \frac{T(D)^h}{\prod_{b=0}^{h-1}(\nu + 2b)} \sum_{\mathcal{P}} \prod_{(s,t)\in\mathcal{P}} H_{j_s j_t}. \tag{29}$$

Here $\mathcal{P}$ runs over the distinct pairings of the $2h$ positions, including their multiplicities when an index repeats. In particular,

$$\langle z_i z_j \rangle = \frac{T H_{ij}}{\nu}, \qquad \langle z_i z_j z_k z_l \rangle = \frac{T^2 (H_{ij}H_{kl} + H_{ik}H_{jl} + H_{il}H_{jk})}{\nu(\nu+2)}. \tag{30}$$

One can first derive these relations after a Wick rotation in an integer dimension, where the angular average is well defined, and then continue the rational tensor coefficients in d. This uses the dimensional-regularization convention of Eq. (1) [1]. Poles in d remain exact rational poles.

For $b_j = -a_j > 0$, expand $\prod_{j:a_j<0}(B_j - 2z_j)^{b_j}$ and insert Eq. (29). The result is a finite polynomial in the positive-sector denominators:

$$\left\langle \prod_{j:a_j<0} D_j^{b_j} \right\rangle = \sum_{\boldsymbol{t}} c_{\boldsymbol{t}}(d) \prod_{i\in S} D_i^{t_i}, \qquad I_{\boldsymbol{a}} = \sum_{\boldsymbol{t}} c_{\boldsymbol{t}}(d) I_{\boldsymbol{a}_S - \boldsymbol{t}}. \tag{31}$$

The vector $\boldsymbol{a}_S$ equals $\boldsymbol{a}$ on S and vanishes elsewhere; the t_i are nonnegative and vanish outside S. The coefficients also depend rationally on the fixed kinematics. Each power $D_i^{t_i}$ cancels the corresponding denominator power. If a cancellation creates a new negative index, that line leaves the positive sector. Repeating the angular step therefore decreases the positive-sector size whenever a numerator remains. A sector with no denominators is scaleless and vanishes in dimensional regularization [1]. The process terminates in scalar sectors with nonnegative indices.

F.3 Lowering powers before removing the numerator

A second exact route starts with IBP. Define $A = \sum_j a_j$ and $Y_{ij} = m_i^2 + m_j^2 - (q_i - q_j)^2$. Expanding the total derivative with vector $\ell - q_i$ gives

$$0 = dI_{\boldsymbol{a}} - \sum_j a_j \int \frac{d^d\ell}{i\pi^{d/2}} \frac{2(\ell - q_i)\cdot(\ell - q_j)}{D_j} \prod_h D_h^{-a_h}, \tag{32}$$

$$2(\ell - q_i)\cdot(\ell - q_j) = D_i + D_j + Y_{ij}, \tag{33}$$

$$0 = (d - A)I_{\boldsymbol{a}} - \sum_j a_j I_{\boldsymbol{a}+\boldsymbol{e}_j-\boldsymbol{e}_i} - \sum_j a_j Y_{ij} I_{\boldsymbol{a}+\boldsymbol{e}_j}. \tag{34}$$

Here $\boldsymbol{e}_j$ is the unit index vector. This derivation holds for signed integer indices, so it includes numerator terms. Let $N_- = \{j : a_j < 0\}$. Moving the negative-index terms to the right gives, for $i \in S$,

$$\sum_{j\in S} a_j Y_{ij} I_{\boldsymbol{a}+\boldsymbol{e}_j} = F_i(\boldsymbol{a}), \tag{35}$$

$$F_i(\boldsymbol{a}) = (d - A)I_{\boldsymbol{a}} - \sum_{j:a_j\neq 0} a_j I_{\boldsymbol{a}+\boldsymbol{e}_j-\boldsymbol{e}_i} - \sum_{j\in N_-} a_j Y_{ij} I_{\boldsymbol{a}+\boldsymbol{e}_j}. \tag{36}$$

For target $\boldsymbol{b}$ with $b_k > 1$, choose $\boldsymbol{a} = \boldsymbol{b} - \boldsymbol{e}_k$. If Y_S is invertible, multiply this system by its inverse to obtain

$$I_{\boldsymbol{b}} = \frac{1}{a_k} \sum_{i\in S} (Y_S^{-1})_{ki} F_i(\boldsymbol{a}). \tag{37}$$

Every integral on the right has a smaller sum of positive indices than $\boldsymbol{b}$, and the numerator rank does not increase. When the positive powers reach one, the angular step removes the remaining numerator. With no negative indices, Eq. (37) reduces to the scalar Cayley recurrence derived earlier. These identities give a finite reduction to unit-power scalar sectors. They produce a spanning set; special kinematics can introduce further relations between its elements.

F.4 Keeping the dimension poles in factored form

The preceding identities restrict the form of every dimension-dependent denominator. The inverse Gram and Cayley matrices have rational entries at the chosen kinematic point, and Eq. (37) introduces only polynomial dependence on d. Equation (29) introduces products of linear factors $d + j$, where j is an integer. By induction through the recurrence, each coefficient therefore has the form

$$r(d) = \frac{P(d)}{\prod_j (d + j)^{n_j}}, \qquad P(d) \in \mathbb{Q}[d], \quad n_j \in \mathbb{Z}_{\geq 0}. \tag{38}$$

The finite products contain only factors introduced by the angular moments. For addition, take Q to be the least common multiple of two such denominators Q_1 and Q_2. Multiplying each summand by its missing factors gives

$$\frac{P_1}{Q_1} + \frac{P_2}{Q_2} = \frac{P_1(Q/Q_1) + P_2(Q/Q_2)}{Q}. \tag{39}$$

The exponent of $d+j$ in Q is the larger of its two exponents. Multiplication adds these exponents. A factor $d+j$ cancels exactly when $P(-j) = 0$, because division with remainder gives $P(d) = (d+j)\widetilde{P}(d) + P(-j)$. We repeat this cancellation for each retained factor. These operations preserve the exact rational function and require only rational polynomial coefficients and integer factor multiplicities. They avoid general rational-function simplification in the new path. The earlier arithmetic checks compare these coefficients with a general rational-function implementation and native Kira reductions. The current application retains the fixed numerator-first recurrence; its new learned decision concerns native seed regions.

G Exact reduction at a zero two-point invariant

A singular Gram matrix prevents the projection used by the generic tensor recurrence. For a two-point sector with $p^2 = 0$, Feynman parameters give a direct reduction to tadpoles. Degenerate one-loop kinematics require such separate identities [7]. We derive the identity used in this study below. It applies to a polynomial numerator and two positive denominator powers; it does not assume that the two masses differ.

G.1 Scalar integrals and mass derivatives

Let x and y denote squared masses, and define

$$T(x) = \int \frac{d^d\ell}{i\pi^{d/2}} \frac{1}{\ell^2 - x + i0}, \qquad B_{ab}(x,y) = \int \frac{d^d\ell}{i\pi^{d/2}} \frac{1}{(\ell^2 - x + i0)^a [(\ell - p)^2 - y + i0]^b}\bigg|_{p^2=0}. \tag{40}$$

Here a, b are positive integers. Dimensional regularization sets $T(0) = 0$. With the normalization in Eq. (1), Wick rotation and the radial beta integral give [1]

$$T(x) = -\Gamma(1 - d/2)(x - i0)^{d/2-1}. \tag{41}$$

For $a = b = 1$, combine the denominators with a parameter z and shift $\ell = k + (1 - z)p$. The combined squared mass is $M^2(z) = zx + (1 - z)y$, because $p^2 = 0$. Integrating over k and then z gives

$$\begin{aligned} B_{11}(x,y) &= \Gamma(2 - d/2) \int_0^1 [zx + (1 - z)y - i0]^{d/2-2}\, dz \\ &= \frac{T(x) - T(y)}{x - y}, \qquad x \neq y. \end{aligned} \tag{42}$$

The second equality follows from the elementary antiderivative of a power and $\Gamma(2 - d/2) = (1 - d/2)\Gamma(1 - d/2)$. Differentiating a propagator with respect to its squared mass raises its power. Therefore

$$B_{ab}(x,y) = \frac{\partial_x^{a-1} \partial_y^{b-1} B_{11}(x,y)}{(a-1)!(b-1)!}. \tag{43}$$

This operation preserves a rational combination of $T(x)$ and $T(y)$ because $\partial_x T(x) = (d/2 - 1)T(x)/x$. We take the derivatives before imposing relations between the masses. If $x = y \neq 0$, the combined mass is independent of z, and the beta integral instead gives

$$B_{ab}(x,x) = T_{a+b}(x), \qquad T_n(x) = \frac{\prod_{j=1}^{n-1}(d/2 - j)}{(n-1)!\, x^{n-1}} T(x). \tag{44}$$

The formula for T_n also follows by differentiating Eq. (41). These identities support independent symbolic squared masses as well as rational values. A relation involving distinct symbolic masses is a generic rational identity; equal masses use Eq. (44) directly.

G.2 Polynomial numerators

Let $P(\ell)$ be a polynomial numerator. Combining the two denominators gives

$$B_{ab}[P] = \frac{\Gamma(A)}{\Gamma(a)\Gamma(b)} \int_0^1 dz\, z^{a-1}(1 - z)^{b-1} \int \frac{d^dk}{i\pi^{d/2}} \frac{P(k + (1 - z)p)}{[k^2 - M^2(z) + i0]^A}, \quad A = a + b. \tag{45}$$

For an auxiliary inverse propagator $(\ell - q_j)^2 - m_j^2$, put $w_j = (1 - z)p - q_j$. Its numerator factor becomes $k^2 + 2k \cdot w_j + w_j^2 - m_j^2$. Both $w_i \cdot w_j$ and $w_j^2 - m_j^2$ are polynomials of degree at most one in z when $p^2 = 0$.

Rotational covariance gives zero for odd moments of k. Tracing two indices of an even moment determines its normalization recursively: if its pairing coefficient is C_h, then $C_h(d+2h-2) = C_{h-1}$ and $C_0 = 1$. Consequently,

$$\left\langle \prod_{j=1}^{2h} (k \cdot w_j) \right\rangle = \frac{(k^2)^h}{\prod_{r=0}^{h-1}(d+2r)} \sum_{\mathcal{P}} \prod_{(i,j)\in\mathcal{P}} w_i \cdot w_j, \tag{46}$$

where $\mathcal{P}$ runs over pairings of the $2h$ positions. The trace recurrence follows because one pairing contributes d and the remaining contractions contribute $2h-2$. This also derives the full-dimensional form of the angular identity used in Appendix F.

After the finite numerator expansion and angular average, define $c_{ht}(d)$ by

$$\langle P(k+(1-z)p)\rangle = \sum_{h,t} c_{ht}(d)(k^2)^h z^t. \tag{47}$$

These coefficients also depend on the fixed external scalar products and numerator masses. They contain no loop momentum. To integrate each term, Wick rotate the radial integral, set $u = k_E^2/M^2$, and use $\int_0^\infty u^{\alpha-1}(1+u)^{-A}du = \Gamma(\alpha)\Gamma(A-\alpha)/\Gamma(A)$ with $\alpha = d/2+h$. This gives

$$J_{Ah}(M^2) = \int \frac{d^dk}{i\pi^{d/2}} \frac{(k^2)^h}{(k^2-M^2+i0)^A} = (-1)^{A-h}\frac{\Gamma(d/2+h)\Gamma(A-h-d/2)}{\Gamma(d/2)\Gamma(A)}(M^2-i0)^{d/2+h-A}. \tag{48}$$

The initial convergence region defines the integrals; analytic continuation gives the dimensional-regularization identity outside that region [1].

For $x \neq y$, set $\Delta = x-y$, $\rho = h-A+1$, and $\lambda = d/2-1$. Expand $(1-z)^{b-1}$ and then substitute $u = y+\Delta z$. For an integer $k \geq 0$,

$$\int_0^1 z^k(y+\Delta z)^{\lambda+\rho}\,dz = \frac{1}{\Delta^{k+1}}\sum_{j=0}^{k}\binom{k}{j}(-y)^{k-j}\frac{x^{\lambda+\rho+j+1}-y^{\lambda+\rho+j+1}}{\lambda+\rho+j+1}. \tag{49}$$

This follows by expanding $(u-y)^k$ and integrating each power of u. Dividing by the tadpole normalization in Eq. (41), define

$$R_h = \frac{(-1)^{A-h+1}(d/2)_h(1-d/2)_{A-h-1}}{(a-1)!(b-1)!}, \qquad (z)_n = \frac{\Gamma(z+n)}{\Gamma(z)}. \tag{50}$$

For a negative integer n, the last ratio means the inverse finite product $1/\prod_{r=1}^{-n}(z-r)$. Equations (45)–(49) now give the explicit rational reduction

$$\begin{aligned} B_{ab}[P] = \sum_{h,t} c_{ht}(d) R_h \sum_{v=0}^{b-1}(-1)^v \binom{b-1}{v} \\ \times \sum_{j=0}^{k}\binom{k}{j}\frac{(-y)^{k-j}}{\Delta^{k+1}}\frac{x^{\rho+j+1}T(x)-y^{\rho+j+1}T(y)}{\lambda+\rho+j+1}, \\ k = a-1+t+v. \end{aligned} \tag{51}$$

If one mass is zero, its endpoint vanishes in a convergence region and hence by analytic continuation; the implementation omits that scaleless endpoint before evaluating powers. If both masses vanish, the sector is scaleless. For equal nonzero masses, use the beta integral directly:

$$B_{ab}[P] = \sum_{h,t} c_{ht}(d) R_h \frac{\Gamma(a+t)\Gamma(b)}{\Gamma(A+t)} x^{h-A+1}T(x). \tag{52}$$

Thus the equal-mass rule has no division by $x-y$. A sector with one positive denominator uses Eq. (48) and the same angular expansion without a Feynman parameter.

At rational kinematics, the coefficients have polynomial numerators in d and products of integer-shift linear factors $d+j$ in their denominators. The finite products in Eqs. (46) and (50), and the linear denominator in Eq. (49), prove this statement. We therefore use the exact factored arithmetic of Appendix F. The implemented tensor domain has at most five coordinates, numerator rank at most four, and at most four extra positive powers. Symbolic masses currently use the scalar derivative identity. Inputs outside these conditions continue through the existing exact rules and Kira.

H Connected one-scale tree reductions

We derive the previously retained tree rules to state precisely which high-loop structures they cover. Let $\mathcal{T}$ be a rooted tree with loop momenta $\ell_0,\dots,\ell_{L-1}$ and parent $\pi(j)$ for every nonroot vertex. A massless two-point family is

$$J_{\mathcal{T}}=\int\prod_{j=0}^{L-1}\frac{d^d\ell_j}{i\pi^{d/2}}\frac{1}{(\ell_0^2)^{a_0}[(\ell_0-p)^2]^c}\prod_{j=1}^{L-1}\frac{1}{(\ell_j^2)^{a_j}[(\ell_j-\ell_{\pi(j)})^2]^{v_j}},\qquad p^2=s\neq 0. \tag{53}$$

The massive vacuum version replaces the two root factors by $(\ell_0^2-m^2)^{-c}$ and has no external momentum. Each edge power v_j, root power c, and leaf self power a_j is positive. Internal self powers can be nonpositive. Auxiliary powers vanish. The implementation admits $L\le 13$ and absolute physical powers at most sixteen. The derivation itself applies to finite trees with meromorphically defined integrals.

Temporarily use negative massless denominators. Feynman parametrization of a massless bubble, translation of its loop momentum, and a radial beta integral give

$$\begin{aligned}G(u,v)&=\frac{\Gamma(u+v-d/2)}{\Gamma(u)\Gamma(v)}\int_0^1 z^{d/2-v-1}(1-z)^{d/2-u-1}\,dz\\&=\frac{\Gamma(u+v-d/2)\Gamma(d/2-u)\Gamma(d/2-v)}{\Gamma(u)\Gamma(v)\Gamma(d-u-v)}.\end{aligned} \tag{54}$$

The remaining scale factor is $(-P^2-i0)^{d/2-u-v}$ for bubble momentum P. The beta identity used here follows by the same radial and parameter integrations as Appendix G; analytic continuation supplies the dimensionally regularized relation [1]. Recursive subintegral integration has established applications to massless propagator integrals [17].

For each nonroot vertex, define

$$u_j=a_j+\sum_{h:\pi(h)=j}\sigma_h,\qquad \sigma_j=u_j+v_j-d/2. \tag{55}$$

Integrating a leaf gives $G(u_j,v_j)$ and transfers power σ_j to its parent. Repeating from the leaves to the root proves the product rule by induction. In the two-point case, set $\alpha=a_0+\sum_{h:\pi(h)=0}\sigma_h$; its dimensionless coefficient is

$$C_{\mathcal{T}}=G(\alpha,c)\prod_{j=1}^{L-1}G(u_j,v_j). \tag{56}$$

The total physical power $A=\sum_j a_j+\sum_{j>0}v_j+c$ gives scale power $(-s-i0)^{Ld/2-A}$. Let $M_{\mathcal{T}}$ set all physical powers to one and let C_0 be its Gamma product. Restoring the original propagator signs and dividing by this master gives

$$J_{\mathcal{T}}=s^{-(A-2L)}\frac{C_{\mathcal{T}}}{C_0}M_{\mathcal{T}}. \tag{57}$$

The sign $(-1)^{A-2L}$ from the propagator convention cancels that from $(-s)^{-(A-2L)}$.

For a massive vacuum root, put $\sigma_0 = \sum_{h:\pi(h)=0} \sigma_h$. Wick rotation and the substitution $x = m^2 t$ give

$$\frac{1}{\Gamma(d/2)} \int_0^\infty \frac{x^{d/2-\sigma_0-1}}{(x+m^2)^c}\, dx = (m^2)^{d/2-\sigma_0-c} \frac{\Gamma(d/2-\sigma_0)\Gamma(\sigma_0+c-d/2)}{\Gamma(d/2)\Gamma(c)}. \tag{58}$$

Multiply the dimensionless Gamma ratio by the nonroot bubble factors to define $C_{\mathcal{T}}^V$. There are $P = 2L-1$ physical lines and $A = c + \sum_{j>0}(a_j + v_j)$. Dividing by the unit-power vacuum master $M_{\mathcal{T}}^V$ yields

$$V_{\mathcal{T}} = (-m^2)^{-(A-P)} \frac{C_{\mathcal{T}}^V}{C_0^V} M_{\mathcal{T}}^V. \tag{59}$$

In either family, a subtree with n_j vertices has $u_j = U_j - (n_j - 1)d/2$ with integer U_j. This follows from Eq. (55) by induction. Target and master Gamma arguments therefore have equal slopes in d and integer differences in their constant terms. Repeated use of $\Gamma(z+1) = z\Gamma(z)$ gives

$$\frac{\Gamma(z+n)}{\Gamma(z)} = \begin{cases} \prod_{k=0}^{n-1}(z+k), & n > 0, \\ 1, & n = 0, \\ [\prod_{k=1}^{-n}(z-k)]^{-1}, & n < 0. \end{cases} \tag{60}$$

Thus the target-to-master coefficients are rational in d and the one scale. Cancelling equal Gamma factors before expanding these finite products preserves the identity and avoids repeated symbolic work. This proves the reductions for the stated trees; additional coupled propagators or auxiliary numerators require another reduction.

I Direct evaluation of vacuum factors

We first identify a case that requires no master integral. If all masses vanish, there is no external momentum, and every active inverse propagator is the square of a homogeneous linear combination of loop momenta, the integrand has no dimensional scale. After Wick rotation, let R be the common radial coordinate for all loop momenta. The measure and denominators give radial power $R^{\omega-1}$ with $\omega = Ld - 2\sum_j a_j$. Split the radial integral at an arbitrary $\mu > 0$ and continue its two pieces separately:

$$\int_0^\mu R^{\omega-1}\, dR + \int_\mu^\infty R^{\omega-1}\, dR = \frac{\mu^\omega}{\omega} - \frac{\mu^\omega}{\omega} = 0. \tag{61}$$

The two pieces initially converge on opposite half-planes and their meromorphic continuations cancel. Angular singularities can be treated by analytic regularization before the same scaling argument. This is the standard scaleless-integral prescription in dimensional regularization [1]. It gives the combined integral as zero; separating ultraviolet and infrared poles requires retaining the two regulated contributions separately. The implementation checks the masses and momentum homogeneity, rejects a zero denominator with positive power, and accepts up to thirteen loops and 128 coordinates. It excludes cut denominators and symbolic-index inputs. Couplings between loop momenta are permitted for this zero-integral rule.

Consider a loop momentum k that occurs only in denominators $k^2 - x_j + i0$, where x_j is a squared mass. There is no external momentum in this component. Denominator powers can be positive, zero, or negative. Equal masses first combine by addition of their powers. With $z = k^2$, let A_x be the resulting integer power at each distinct squared mass x. The integrand is the rational function

$$F(z) = \prod_x (z-x)^{-A_x}. \tag{62}$$

For a mass with $A_x > 0$, multiply by $(z-x)^{A_x}$ and expand the remaining analytic factor at $z = x$. Its first A_x Taylor coefficients give the principal part at that pole. Subtracting every principal

part leaves a rational function without a finite pole, which is a polynomial $P(z)$. Thus

$$F(z) = P(z) + \sum_{x:A_x>0} \sum_{r=1}^{A_x} \frac{c_{xr}}{(z-x)^r}, \tag{63}$$

$$c_{xr} = \frac{1}{(A_x-r)!} \frac{d^{A_x-r}}{dz^{A_x-r}} \prod_{y\neq x} (z-y)^{-A_y} \bigg|_{z=x}. \tag{64}$$

This derivation also covers negative powers, since they supply polynomial factors in the product. For distinct rational masses, the coefficients are rational numbers. The implementation obtains them by multiplying truncated Taylor series, so it avoids general symbolic differentiation. A factor with exponent $-A_y$ has coefficient

$$[t^n](x-y+t)^{-A_y} = \binom{-A_y}{n} (x-y)^{-A_y-n}. \tag{65}$$

The binomial series proves Eq. (65); only $0 \le n < A_x$ is needed in Eq. (64). Equal masses must be combined before this expansion, which avoids division by a vanishing mass difference.

Polynomial terms in k and massless vacuum integrals have no scale and vanish in dimensional regularization [1]. For a nonzero squared mass x, define the normalized tadpole

$$T_r(x) = \int \frac{d^dk}{i\pi^{d/2}} (k^2 - x + i0)^{-r}. \tag{66}$$

To evaluate it, first take $x > 0$ and a convergent Euclidean domain. The Schwinger representation follows from the defining Gamma integral after the change of variable $u = \alpha(q^2 + x)$:

$$(q^2+x)^{-r} = \frac{1}{\Gamma(r)} \int_0^\infty \alpha^{r-1} e^{-\alpha(q^2+x)}\, d\alpha. \tag{67}$$

The Gaussian momentum integral is $\int d^dq\, e^{-\alpha q^2} = \pi^{d/2}\alpha^{-d/2}$. Inserting this in Eq. (67) and setting $u = \alpha x$ gives $x^{d/2-r}\Gamma(r-d/2)/\Gamma(r)$. Wick rotation supplies $(-1)^r$ for the denominator convention in Eq. (66). Analytic continuation therefore gives

$$T_r(x) = (-1)^r \frac{\Gamma(r-d/2)}{\Gamma(r)} (x-i0)^{d/2-r}, \qquad \frac{T_r(x)}{T_1(x)} = \frac{\prod_{j=1}^{r-1}(d/2-j)}{(r-1)!x^{r-1}}. \tag{68}$$

The second equality follows by applying $\Gamma(z+1) = z\Gamma(z)$ repeatedly to the first. The empty product at $r = 1$ is one. Applying Eq. (68) term by term in Eq. (63) proves both the reduction to unit tadpoles and their full analytic evaluation. The result is a meromorphic expression in d; ultraviolet or infrared poles are retained. The same tadpole value also evaluates any unit tadpole returned by the retained zero-invariant tensor formulas. A remaining master with several denominators still requires its own evaluation.

Independent components multiply, as proved in Appendix A. The direct implementation accepts one through thirteen such loop momenta, at most eight denominators per component, individual absolute powers at most eight, and total positive power after equal-mass combination at most 48 per component. Masses are nonnegative rational numbers. Denominators with nonzero target power must carry one of those individual momenta with coefficient $+1$ or -1. These tests exclude couplings between components and specify the actual scope of the rule.

The connected one-mass vacuum trees have a separate analytic evaluation. Appendix H derives their dimensionless Gamma product $C_{\mathcal{T}}^V$. With total physical power A, Wick rotation of all integer-power denominators gives

$$V_{\mathcal{T}} = (-1)^A (m^2 - i0)^{Ld/2-A} C_{\mathcal{T}}^V. \tag{69}$$

The exponent follows either from the final radial integral or from the mass dimension of the L measures and A denominators. For a unit-power master, $A = P = 2L - 1$. The application now returns its explicit Gamma arguments and scale exponent together with the rational reduction. This exposes the analytic master value already implied by the tree derivation and avoids a separate master evaluation for these families.

J Analytic inputs and checks

The two-mass vacuum grid contains one request at each $L = 1, \ldots, 13$. At loop order L, each independent component has squared masses $(h+2, h+7)$ with $h = 23011+L$. Its four final target power pairs are $(2,1), (3,1), (2,3), (3,-1)$, and each full target repeats the same pair across all L components. These are products of one-loop factors. The connected vacuum grid has 34 requests: chains at $L = 1, \ldots, 13$, stars at $L = 3, \ldots, 13$, and binary trees at $L = 4, \ldots, 13$. Their parent maps are $\pi(j) = j - 1$, $\pi(j) = 0$, and $\pi(j) = \lfloor (j-1)/2 \rfloor$, respectively. These starting orders omit coincident small-loop routings. The root squared mass remains symbolic. Each final request has root power two, root power three, root and final edge powers two, and root power four, with every other physical power one. The one-loop request instead uses powers two through five. The unit-power integral remains the master. The profiles are $23111 + L$, $23211 + L$, and $23311 + L$ for the three shapes and fix any auxiliary construction choices. These requests test preservation of the connected identity and the earlier symbolic single-loop dispatch.

The repeated-mass coverage grid uses four denominators per component with squared masses $(h+2, h+7, h+2, 0)$, where $h = 23511+L$. The four power assignments are $(1,1,2,0)$, $(2,1,1,1)$, $(1,2,0,-2)$, and $(0,0,0,2)$. This tests repeated masses, a massless pole, a polynomial numerator, and a scaleless target. It is checked through all thirteen loop orders, with no speed claim against an unfinished or unsupported native calculation.

The scaleless coverage grid has one family for each $L = 1, \ldots, 13$, with inverse propagators ℓ_i^2 and $(\ell_i - \ell_j)^2$ for all $i < j$. It therefore contains $L(L+1)/2$ coordinates and coupled loop momenta for $L > 1$. The three targets set all powers to one, raise the first power to two, or change the first power to minus one. Equation (61) supplies the reference zero. Native one- and two-loop checks test the same rule separately.

The three retained zero-invariant checks use the unequal, equal, and mixed mass choices of Eq. (17), with profiles $23211, 23212, 23213$, numerator rank two, and two excess positive powers. The second excess power is placed on the first physical denominator. The three ordinary tensor checks use Eq. (18) with $h = 23311 + P$ for $P = 2, 3, 4$. Their numerator rank and excess positive power are both two, with one of the excess powers placed on the first physical denominator.

The new native reconstruction inputs are specified in Appendix K.

An independent symbolic partial-fraction calculation checks the vacuum grid. Random rational-function checks verify that subtracting the proposed poles leaves only a polynomial. Convergent Euclidean radial quadrature checks the tadpole normalization, and a native IBP comparison checks its raised-power recurrence. Native box and ordinary tensor coefficients are checked against independently prepared Kira reductions. The connected tree comparison checks preservation against the preceding exact rule; its mathematical justification is the separate derivation in Appendix H.

K Native reconstruction inputs

The planar and nonplanar boxes use Eqs. (19) and (20), respectively. The high-rank targets are $(1,1,1,1,1,1,1,-u,-v)$ with $u, v \geq 0$ and $1 \leq u + v \leq 4$. Each graph contributes fourteen training, six validation, and six reserved high-rank batches. The complete target sets differ

between splits and exclude the preceding comparison batches. Individual target vectors can occur in more than one batch.

For batch index $i = 0, \ldots, 25$, define $j = i \bmod 6$ and $q = \lfloor i/6 \rfloor$. Let

$$\begin{aligned}(A_0, \ldots, A_5) &= (3, 13, 103, 1009, 10007, 100003),\\ (B_0, \ldots, B_5) &= (7, 19, 107, 1013, 10009, 100019).\end{aligned} \tag{70}$$

We set $s = -(A_j + 2q)$ and $t = -(B_j + 4q)$ in both graphs, using each graph's scalar-product convention. Indices 0–13 train the model, 14–19 select it, and 20–25 test it. These equations define the chosen input points. In the planar graph the fourth momentum is $-p_1 - p_2 - p_3$, and in the nonplanar graph it is $p_1 + p_2 - p_3$. Expanding their squares with the listed scalar products gives zero in both cases. The test batches are specified in Table 10.

Table 10: Reserved high-rank target batches. The same (u, v) sets and invariants enter both box routings. Every row defines a complete request for each graph.

i	s	t	Numerator powers (u, v)
20	-109	-119	$(2,0), (0,2), (0,1)$
21	-1015	-1025	$(2,2), (0,4)$
22	-10013	-10021	$(3,1), (3,0), (2,1), (1,3), (0,1)$
23	-100009	-100031	$(3,1), (2,2), (2,1), (2,0), (1,2), (0,4)$
24	-11	-23	$(2,2), (2,1), (1,2), (1,1), (0,2), (0,1)$
25	-21	-35	$(2,1), (1,3), (1,2), (1,0), (0,2), (0,1)$

For the lower-rank box requests, the sector corner is $(1, 1, 1, 1, 1, 1, 1, 0, 0)$. A numerator request uses the supplied first-rank auxiliary targets, and a raised-power request increases one physical index. Table 11 lists their precise target vectors. Their profiles are $h = 27701, 27713$ for training, 27733 for validation, and 27739 for testing, with the low-rank kinematics of Appendix D.

The sunrise family has loop momenta (ℓ_1, ℓ_2) and one external momentum p. Its inverse-propagator momenta and squared masses are

$$(Q_1, \ldots, Q_5) = (\ell_1, \ell_2, p - \ell_1 - \ell_2, p - \ell_1, p - \ell_2), \qquad (m_1^2, \ldots, m_5^2) = (3, 7, 13, 0, 0). \tag{71}$$

The first three coordinates are physical denominators and the last two are auxiliary. The three-loop vacuum uses

$$\begin{aligned}(Q_1, \ldots, Q_6) &= (\ell_1, \ell_2, \ell_3, \ell_1 - \ell_2, \ell_2 - \ell_3, \ell_1 - \ell_3),\\ (m_1^2, \ldots, m_6^2) &= (1, 1, 1, 13, 13, 2h + 53).\end{aligned} \tag{72}$$

All six coordinates are physical. Both structures use the same profile split as the low-rank boxes. For the sunrise, $p^2 = -1009 - 12h$. These assignments specify test families, rather than universal relations between masses and invariants. In every native request the kinematic quantities are fixed rational numbers and d remains symbolic, so the coefficients belong to $\mathbb{Q}(d)$. This test does not reconstruct general symbolic scattering invariants.

The protected native bases consist of the integral indices below. Each listed vector $\boldsymbol{b}$ denotes $M_{\boldsymbol{b}} = I_{\boldsymbol{b}}$ in its corresponding family. The independently prepared symbolic and reconstructed calculations use the same basis for each request. The lists specify the comparison basis; they do not assert a proof of global master independence from a finite seed system.

Planar box:

$$\begin{aligned}\{&(0,0,1,1,0,0,1,0,0), (0,1,0,0,0,1,1,0,0),\\ &(0,1,0,1,1,0,1,0,0), (0,1,1,1,0,1,1,0,0),\\ &(1,0,1,1,1,0,0,0,0), (1,1,1,0,0,1,1,0,0),\\ &(1,1,1,1,1,1,1,-1,0), (1,1,1,1,1,1,1,0,0)\}.\end{aligned}$$

Table 11: Lower-rank reserved targets. Training and validation use the same index patterns at different physical points. A corner request contains the all-unit physical sector.

Structure and request	Target indices
Planar box, corner	$(1,1,1,1,1,1,1,0,0)$
Planar box, numerator	$(1,1,1,1,1,1,1,0,-1),(1,1,1,1,1,1,1,-1,0)$
Planar box, raised power	$(1,1,1,1,1,2,1,0,0),(1,1,1,1,1,1,2,0,0),(2,1,1,1,1,1,1,0,0)$
Nonplanar box, corner	$(1,1,1,1,1,1,1,0,0)$
Nonplanar box, numerator	$(1,1,1,1,1,1,1,0,-1),(1,1,1,1,1,1,1,-1,0)$
Nonplanar box, raised power	$(1,1,1,1,1,2,1,0,0),(1,1,1,1,1,1,2,0,0),(2,1,1,1,1,1,1,0,0)$
Sunrise, corner	$(1,1,1,0,0)$
Sunrise, raised power	$(1,2,1,0,0),(1,1,2,0,0),(2,1,1,0,0)$
Vacuum, corner	$(1,1,1,1,1,1)$
Vacuum, raised power	$(1,2,1,1,1,1),(1,1,2,1,1,1),(1,1,1,2,1,1)$

Nonplanar box:

$$\{(0,0,1,1,1,1,0,0,0),(0,1,0,0,1,1,0,0,0),$$
$$(0,1,1,1,1,1,1,0,0),(1,0,0,1,0,1,0,0,0),$$
$$(1,0,1,1,1,1,0,0,0),(1,1,0,0,0,0,1,0,0),$$
$$(1,1,0,1,0,1,1,0,0),(1,1,0,1,1,1,0,0,0),$$
$$(1,1,1,0,1,0,1,0,0),(1,1,1,1,0,0,1,0,0),$$
$$(1,1,1,1,1,1,1,-1,0),(1,1,1,1,1,1,1,0,0)\}.$$

Sunrise, basis 1:

$$\{(0,1,1,0,0),(1,0,1,0,0),$$
$$(1,1,0,0,0),(1,1,1,-1,0),$$
$$(1,1,1,0,-1),(1,1,1,0,0),$$
$$(1,1,2,0,0)\}.$$

Sunrise, basis 2:

$$\{(1,1,1,0,0)\}.$$

Three-loop vacuum, basis 1:

$$\{(0,1,1,1,0,1),(0,1,1,1,0,2),$$
$$(0,1,1,2,0,1),(1,0,0,1,1,1),$$
$$(1,0,1,1,0,1),(1,0,1,1,1,0),$$
$$(1,0,1,1,1,1),(1,0,1,1,2,0),$$
$$(1,1,0,1,0,1),(1,1,0,1,1,0),$$
$$(1,1,0,1,1,1),(1,1,1,0,0,1),$$
$$(1,1,1,1,0,0),(1,1,1,1,0,1),$$
$$(1,1,1,1,1,0),(1,1,1,1,1,1)\}.$$

Three-loop vacuum, basis 2:

$$\{(1,1,1,1,1,1)\}.$$

L Coefficient reconstruction

Kira and FireFly provide the finite-field algorithms used here [10, 12, 13]. To explain the relation to an exact rational coefficient, suppose

$$r(d) = P(d)/Q(d), \qquad P(d) = \sum_{j=0}^{n} p_j d^j, \quad Q(d) = \sum_{j=0}^{m} q_j d^j. \tag{73}$$

One nonzero denominator coefficient fixes the overall normalization. At a nonsingular point $d = z$ modulo a prime, a numerical reduction returns $r(z)$. Multiplying by the denominator gives the linear constraint

$$\sum_{j=0}^{n} p_j z^j - r(z) \sum_{j=0}^{m} q_j z^j = 0 \pmod{p}. \tag{74}$$

This follows directly from Eq. (73). With correct degree bounds and sufficiently many independent nonsingular samples, the constraints determine the normalized coefficients over the field. Combining suitable primes and reconstructing rational numbers then supplies rational coefficients [12, 13]. FireFly determines the necessary information adaptively; our model chooses the native procedure and probe batch size.

The field operations are exact, while reconstruction termination relies on its degree discovery and checks. Our evaluation therefore also subtracts every reconstructed coefficient from a separately computed symbolic reference and requires the rational difference to vanish. This checks the actual tested answer. Neither a model score nor a successful reduction to the supplied basis alone proves every coefficient for an arbitrary unseen request.

M Exact reduction of a one-scale massless triangle

The on-shell triangle has a Gamma-function representation [11]. We derive its coefficient in our normalization to specify the signs and the master basis used by the implementation. Let

$$T_{abc}(Q, d) = \int \frac{d^d k}{i\pi^{d/2}} \frac{1}{[k^2 + i0]^a [(k - p_1)^2 + i0]^b [(k - p_1 - p_2)^2 + i0]^c}, \tag{75}$$

where $p_1^2 = p_2^2 = 0$, $Q = -(p_1 + p_2)^2 > 0$, and a, b, c are propagator powers. Write $A = a + b + c$ and $h = d/2$. For initially positive powers in a convergence region, Feynman parametrization and the shifted loop integral give

$$T_{abc} = \frac{(-1)^A \Gamma(A - h)}{\Gamma(a)\Gamma(b)\Gamma(c)} \int_{x_j \geq 0} d^3x\, \delta(1 - x_1 - x_2 - x_3) x_1^{a-1} x_2^{b-1} x_3^{c-1} (Q x_1 x_3)^{h-A}. \tag{76}$$

The only nonzero squared momentum difference is between the first and third propagators, which gives the factor $Q x_1 x_3$. The Dirichlet integral follows by writing three Gamma integrals and changing their variables to a common scale and three fractions. Explicitly, the substitution $t_j = u x_j$ in $\prod_j \int_0^\infty dt_j\, t_j^{z_j - 1} e^{-t_j}$ gives

$$\int d^3x\, \delta(1 - \textstyle\sum_j x_j) \prod_j x_j^{z_j - 1} = \frac{\prod_j \Gamma(z_j)}{\Gamma(\sum_j z_j)}. \tag{77}$$

Applying this identity with $z_1 = h - b - c$, $z_2 = b$, and $z_3 = h - a - b$ yields

$$T_{abc} = (-1)^A Q^{h-A} \frac{\Gamma(A - h)\Gamma(h - b - c)\Gamma(h - a - b)}{\Gamma(a)\Gamma(c)\Gamma(d - A)}. \tag{78}$$

Dimensional continuation defines the expression outside the initial convergence region. The limit $b \to 0$ gives the two-point integral obtained by removing the middle propagator. Its unit-power value is

$$B(Q,d) = T_{101} = Q^{h-2}\frac{\Gamma(2-h)\Gamma(h-1)^2}{\Gamma(d-2)}. \tag{79}$$

For positive integers a, c and a nonnegative integer b, division of Eq. (78) by Eq. (79) gives a rational coefficient. To evaluate it with finite products, define

$$(z)_n = \frac{\Gamma(z+n)}{\Gamma(z)} = \begin{cases} \prod_{j=0}^{n-1}(z+j), & n \geq 0, \\ \dfrac{1}{\prod_{j=1}^{-n}(z-j)}, & n < 0. \end{cases} \tag{80}$$

The empty product equals one. The identity follows by repeated application of $\Gamma(z+1) = z\Gamma(z)$ and gives

$$T_{abc} = R_{abc}(Q,d)B(Q,d), \tag{81}$$

$$R_{abc} = \frac{(-1)^A Q^{2-A}}{(a-1)!(c-1)!}(2-h)_{A-2}(h-1)_{1-a-b}(h-1)_{1-b-c}(d-A)_{A-2}. \tag{82}$$

For example, $R_{111} = 2(d-3)/[Q(d-4)]$. Increasing a by one gives the recurrence

$$R_{a+1,b,c} = -\frac{(A-h)(d-A-1)}{Qa(h-a-b-1)}R_{abc}, \tag{83}$$

which supplies a separate exact check on the finite-product calculation. If $a = 0$ or $c = 0$, the remaining massless denominators have no nonzero external invariant. Those sectors vanish in dimensional regularization [1].

The implementation checks all masses and pairwise momentum differences before using this formula. It accepts rational $Q > 0$, powers from zero through four, and $A \leq 8$. It rejects cut propagators, additional occupied lines, and negative indices. For independent loop components with no occupied cross-loop denominator, the product measure then gives

$$\prod_{j=1}^{L} T_{a_j b_j c_j}(Q_j, d) = \left[\prod_{j=1}^{L} R_{a_j b_j c_j}(Q_j, d)\right] \prod_{j=1}^{L} B(Q_j, d). \tag{84}$$

The proof follows by factoring the integrand and measure in a common convergence region and then continuing the result analytically. If the requested integer powers have no common convergence region in d, we first allow the propagator powers to vary independently. Equation (78) gives their meromorphic continuation, which defines the product at the requested powers and dimension. This extension covers products through thirteen loops in the implementation. The implementation retains this product of component coefficients instead of expanding it into a larger polynomial. It supplies no formula for a general connected thirteen-loop scattering graph.

M.1 Inputs for the additional comparison

The four native propagator structures are those in Appendix K. For the new planar and nonplanar box requests, the invariants are $s = -(137+6i)$ and $t = -(167+4i)$, with the incoming/outgoing momentum conventions stated there. Training uses $i = 2, \ldots, 13$, validation uses $i = 14, \ldots, 18$, and reserved tests use $i = 19, \ldots, 23$. Each box target has powers $(1,1,1,1,1,1,1,-a,-b)$. The reserved pairs are listed below; both box structures use the same pairs at each i.

i	Pairs (a, b) in the target request
19	$(3,0),(2,2),(0,2)$
20	$(4,0),(3,1)$
21	$(4,0),(1,2),(0,2),(0,1)$
22	$(3,1),(2,1),(0,2)$
23	$(2,1),(2,0),(1,0),(0,4),(0,2),(0,1)$

The sunrise masses remain $(3, 7, 13)$ and its external invariant is $p^2 = -386221 - 12i$. The three-loop vacuum masses remain $(1, 1, 1, 13, 13, 64255 + 2i)$. These families use the same index splits. Even i requests the sector corner, and odd i requests the raised-power targets defined for that structure. Full physical inputs differ between splits. Graph structures and individual index vectors recur; box target batches are distinct across the new splits.

For a triangle product with L loops, every factor has $Q = 101 + 2L$. We choose factor powers from $(1,1,1)$, $(2,1,1)$, $(1,2,1)$, $(1,1,2)$, $(2,2,1)$, $(1,0,1)$, and $(2,0,2)$. We form six products per loop order and remove duplicate full targets. Cross-loop auxiliary powers are zero. Seven additional single-loop requests use $Q = 211 + 2j$, $j = 1, \ldots, 7$, and the targets $(1,1,1)$, $(1 + (j \bmod 3), 1 + (j \bmod 2), 1)$, $(1,0,1)$, $(2,0,1)$, and $(0,1,1)$, again with duplicate targets removed.

The preceding triangle basis uses unit-power triangle masters. For verification, each such master is replaced by $2(d-3)B/[Q(d-4)]$, and independent components multiply. Kira provides a separate bubble-basis reference for each new single-loop request. The higher-loop comparison combines this exact basis conversion with the product derivation and independent Gamma-function checks.

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